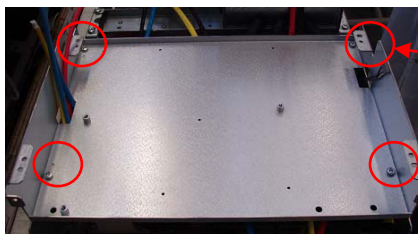


Power Board KIT: VX5A1HD37Y

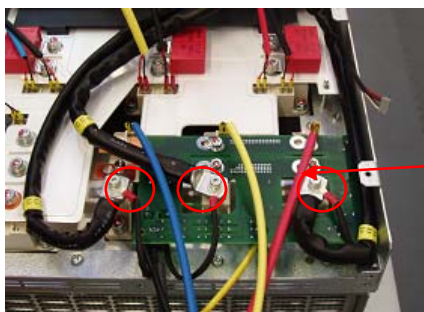


Remove 4 screws (S03).

The power insulate sheet and the PCBA chassis can be removed.

	Mark	Size	Torque
	S03	M4x10	2Nm

Screws KIT: VY1ADV1116

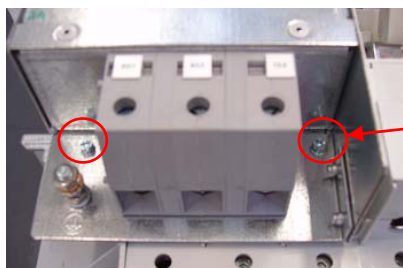


Remove 3 screws (S08).

S08 :
3,5Nm (M6x12) for 75 and 90 kW

	Mark	Size	Torque
	S08	M5x20	2,5Nm

EMC Filter: VX4A1122



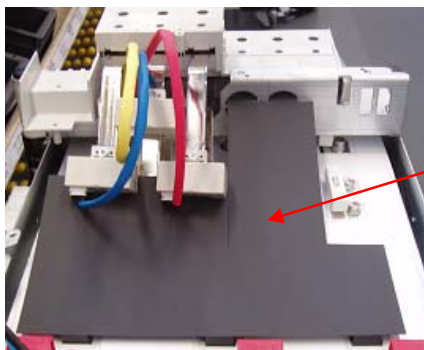
Remove 2 screws (S07).



Remove 4 screws (S03).

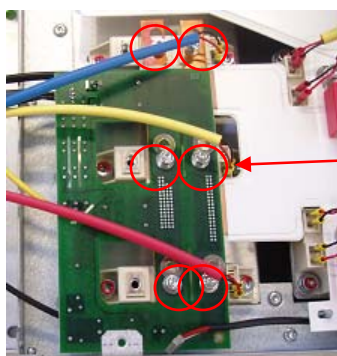
	Mark	Size	Torque
	S07	M4x10	2Nm
	S03	M4x10	1,5Nm

Power Board KIT: VX5A1HD37Y



Remove the insulate sheet power.

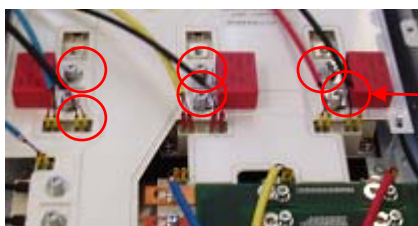
Snubber board: VX4A1208



Remove 6 screws (S11).

	Mark	Size	Torque
	S11	M5x12	2,5Nm

Screws KIT: VY1ADV1116

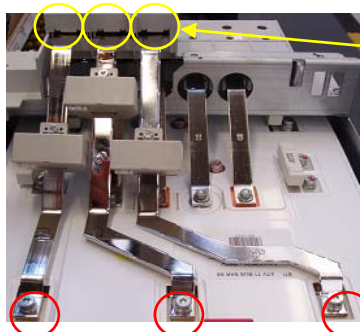


Remove 6 screws (S05).

	Mark	Size	Torque
	S05	M6x12	3,5Nm

Current Sensor: VY1A1111

Inter KIT: VZ3N1353

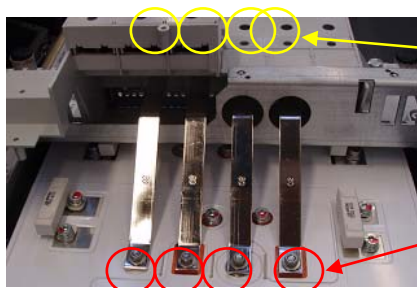


Unscrew 3 screws.

Remove 6 screws (S05).

	Mark	Size	Torque
	S05	M6x12	3,5Nm

Inter KIT: VZ3N1353

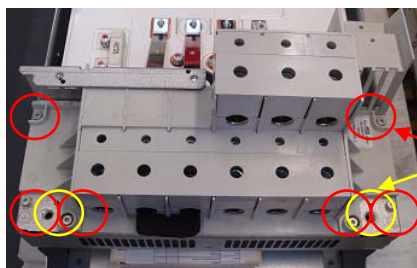


Unscrew 4 screws.

Remove 4 nuts (S05).

	Mark	Size	Torque
	S05	M5x8	2,5Nm

Power Terminal: VZ3N1209



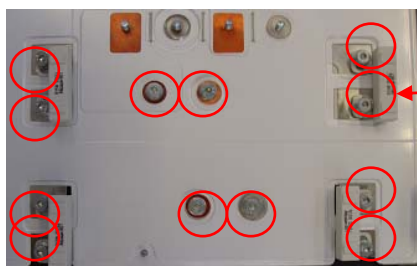
Remove 2 screws (S09).

Remove 6 screws (S22).

	Mark	Size	Torque
	S09	M8x20	1,5Nm
	S22	M5x25	2,5Nm

Lot of 4 capacitors: VY1ADC1202

Bus Bar: VZ3N1354

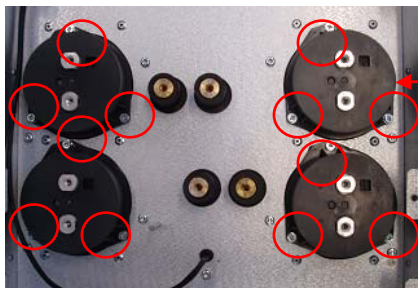


Remove 12 screws (S05).

The bus bar can be removed.

	Mark	Size	Torque
	S05	M6x12	3,5Nm

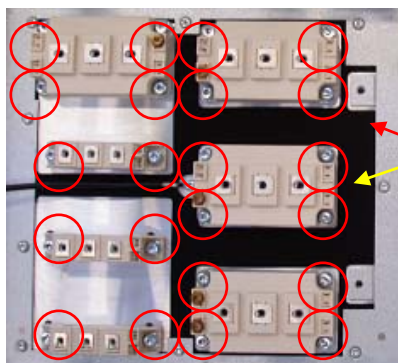
Lot of 4 capacitors: VY1ADC1202



Remove 12 screws (S07).

	Mark	Size	Torque
	S07	M4x10	0,78Nm

Power Modules KIT: VZ3IM2200M3745Y

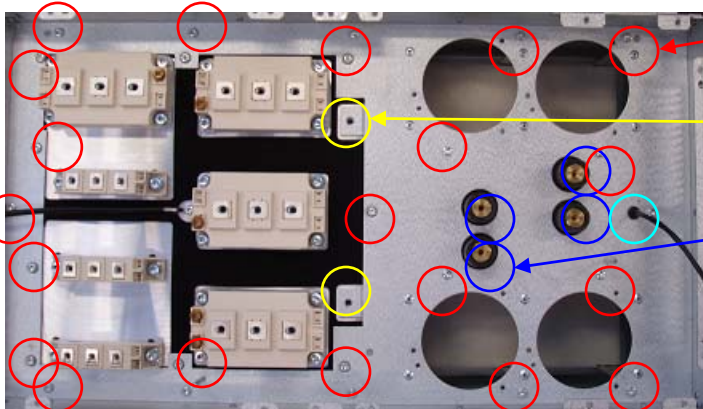


Remove 22 screws (S06).

Remove 6 screws (S22).

	Mark	Size	Torque
	S06	M6x20	4,4Nm

Screws KIT: VY1ADV1115



Remove 20 screws (S03).

Remove 2 screws (S05).

Remove 4 screws (S04).

Remove the cable (E103).

The chassis can be removed.

	Mark	Size	Torque
	S03	M4x10	2Nm
	S05	M6x12	3,5Nm
	S04	4x12	2Nm

Fan KIT: VZ3V1208



Remove this screw (S03).



Disconnect 2 wires.

	Mark	Size	Torque
	S03	M4x10	2Nm

9.20.2 Product Assembling Drawing

Refer to following file: [Assembling Cabling aav3467900-4.1.pdf](#)

9.20.3 Product Cabling Drawing

Refer to following file:



9.21 ATV61/71 Size 9 (size, refer to 1.2)

9.21.1 Dismantling and reassembling

Size 9: ATV71HD90N4, ATV71HD55M3, ATV61HD90N4, ATV61HC11N4, ATV61HD55M3, ATV61HD75M3

ATV71HD90N4	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1215	Fan KIT
VZ3V1214	Internal Fan
VZ3TM1400M1271	Braking Module 400A / 1200V
VZ3S1910	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1327	DC BUS BAR KIT
VZ3N1325	Kit BUS / Bars AC
VZ3N1323	Wires KIT
VZ3IM1400M1271	Module IGBT 400A / 1200V
VZ3G1102	Thermal Sensor
VY1ADV1105	Screws KIT
VY1ADC1110	Lots of 4 capacitors DC bus 6800µF / 400V
VY1A1404	Plastic KIT
VY1A1303	Assembling KIT
VY1A1210	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD9011	Power Board
VX5A1300	Soft Charge Board
VX4A71101Y	Control bloc P >=90kW
VX4A1116	RFI Filter Board
VW3A4521	DC Choke
VZ3TD1250M1671	Rectifier Module (Thyristor / Diode) 250A / 1600V

ATV71HD55M3	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1215	Fan KIT
VZ3V1214	Internal Fan
VZ3TM1400M0671	Braking Module IGBT 400A / 600V
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3S1911	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1337	DC BUS BAR KIT
VZ3N1335	Kit BUS / Bars AC
VZ3N1323	Wires KIT
VZ3IM1600M0671	Module IGBT 600A / 600V
VZ3G1102	Thermal Sensor
VY1ADV1105	Screws KIT
VY1ADC1115	Lots of 3 capacitors DC bus 6800µF / 400V
VY1A1404	Plastic KIT
VY1A1303	Assembling KIT
VY1A1210	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD5575	Power Board
VX5A1300	Soft Charge Board
VX4A71101Y	Control bloc P >=90kW
VX4A1116	RFI Filter Board
VW3A4524	DC Choke

ATV61HD90N4	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1215	Fan KIT
VZ3V1214	Internal Fan
VZ3TM1400M1271	Braking Module 400A / 1200V
VZ3TD1250M1671	Rectifier Module (Thyristor / Diode) 250A / 1600V
VZ3S1910	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1327	DC BUS BAR KIT
VZ3N1325	Kit BUS / Bars AC
VZ3N1323	Wires KIT
VZ3IM1400M1271	Module IGBT 400A / 1200V
VZ3G1101	Thermal Sensor
VY1ADV1105	Screws KIT
VY1ADC1110	Lots of 4 capacitors DC bus 6800µF / 400V
VY1A1404	Plastic KIT
VY1A1303	Assembling KIT
VY1A1210	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD9011	Power Board
VX5A1300	Soft Charge Board
VX4A61101Y	Control board P >= 90kW
VX4A1116	RFI Filter Board
VW3A4521	DC Choke
VZ3G1102	Thermal Sensor

ATV61HC11N4	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits
VZ3V1215	Fan KIT
VZ3V1214	Internal Fan
VZ3TM1400M1271	Braking Module 400A / 1200V
VZ3TD1250M1671	Rectifier Module (Thyristor / Diode) 250A / 1600V
VZ3S1910	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1327	DC BUS BAR KIT
VZ3N1325	Kit BUS / Bars AC
VZ3N1323	Wires KIT
VZ3IM1400M1271	Module IGBT 400A / 1200V
VZ3G1101	Thermal Sensor
VY1ADV1105	Screws KIT
VY1ADC1110	Lots of 4 capacitors DC bus 6800µF / 400V
VY1A1404	Plastic Parts KIT
VY1A1303	Assembling KIT
VY1A1210	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD9011	Power Board
VX5A1300	Soft Charge Board
VX4A61101Y	Control board P >= 90kW
VX4A1116	RFI Filter Board
VW3A4521	DC Choke
VZ3G1102	Thermal Sensor

ATV61HD55M3	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1215	Fan KIT
VZ3V1214	Internal Fan
VZ3TM1400M0671	Braking Module IGBT 400A / 600V
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3S1911	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1337	DC BUS BAR KIT
VZ3N1335	Kit BUS / Bars AC
VZ3N1323	Wires KIT
VZ3IM1600M0671	Module IGBT 600A / 600V
VZ3G1102	Thermal Sensor
VY1ADV1105	Screws KIT
VY1ADC1115	Lots of 3 capacitors DC bus 6800µF / 400V
VY1A1404	Plastic KIT
VY1A1303	Assembling KIT
VY1A1210	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD5575	Power Board
VX5A1300	Soft Charge Board
VX4A61101Y	Control board P >= 90kW
VX4A1116	RFI Filter Board
VW3A4524	DC Choke

ATV61HD75M3	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1214	Internal Fan
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3S1911	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3IM1600M0671	Module IGBT 600A / 600V
VZ3G1102	Thermal Sensor
VY1ADC1116	Lots of 4 capacitors DC bus 6800µF / 400V
VY1A1211	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD7590	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX4A61101Y	Control board P >= 90kW
VX4A1116	RFI Filter Board
VW3A4525	DC Choke
VY1A1404	Plastic KIT
VZ3N1323	Wires KIT
VZ3N1335	Kit BUS / Bars AC
VZ3N1337	DC BUS BAR KIT
VZ3V1215	Fan KIT
VY1A1303	Assembling KIT
VZ3TM1400M0671	Braking Module IGBT 400A / 600V
VY1A1210	Front Cover incl.I/O Terminal Cover

Front Cover with I/O cover: VY1A1210



Remove 7 screws



The Front Cover can be removed.

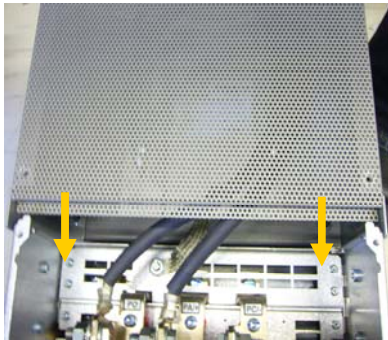


Size	Torque
M6x10	4.4Nm

DC Choke Front Cover



Remove 4 screws

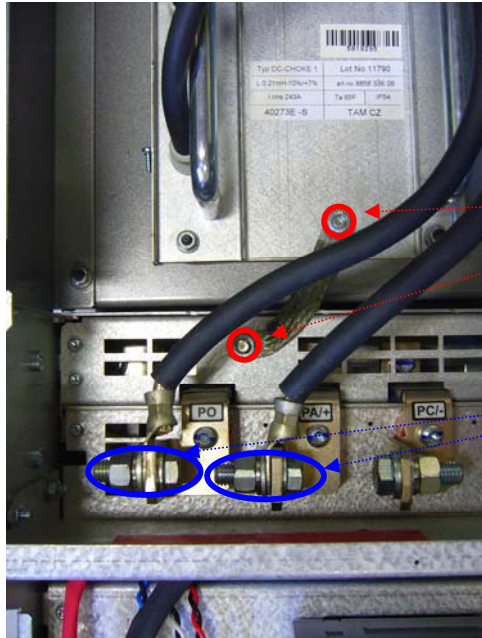


The DC Choke front cover can be removed.



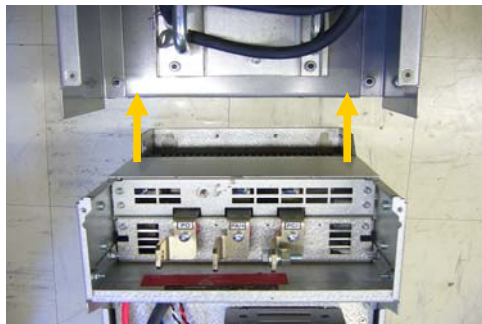
Size	Torque
M6x12	4.4Nm

DC Choke connection



Remove 2 nuts (P1).

Remove 2 screws (P2).

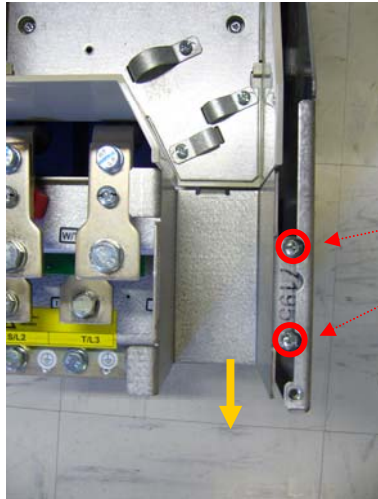


The DC Choke can be removed.



Mark	Size	Torque
P1	M8	10,8Nm
P2	M12x35	45Nm

Metal Part KIT:VY1A1303



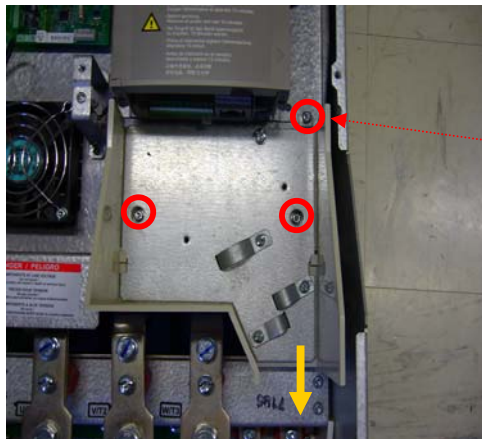
Remove 2 screws

The metal part can be removed.



Size	Torque
M4x10	1.2Nm

Plastic part KIT: VY1A1404



Remove 3 screws

The Plastic Part KIT can be removed.



Size	Torque
M4x10	1.2Nm

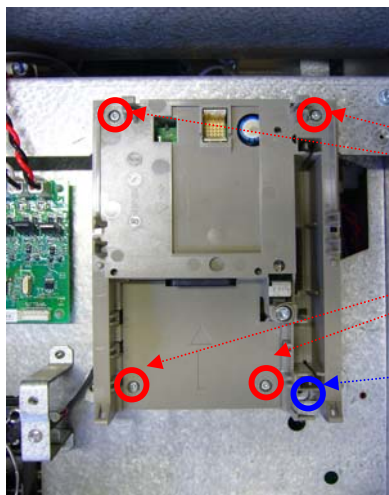
Control bloc: VX4A71101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board



Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 – S8).

Remove 1 screw (S38).



Disconnect the ribbon cable.
X4 from Control bloc" interface Board"->CNX4 on Power Board

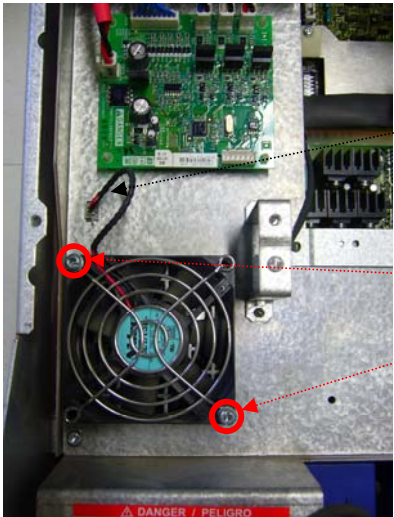
Disconnect the ribbon cable.
X3 from Control bloc "interface Board"->X3 on Motor Control Board

The Control bloc can be removed.

Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5Nm



Internal Fan: VZ3V1214



Disconnect the wire

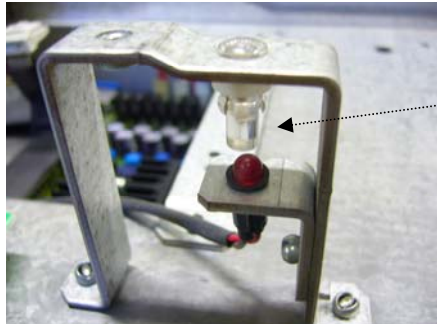
Remove 2 screws

The Internal Fan can be removed.

Size	Torque
M4x35	1.2Nm



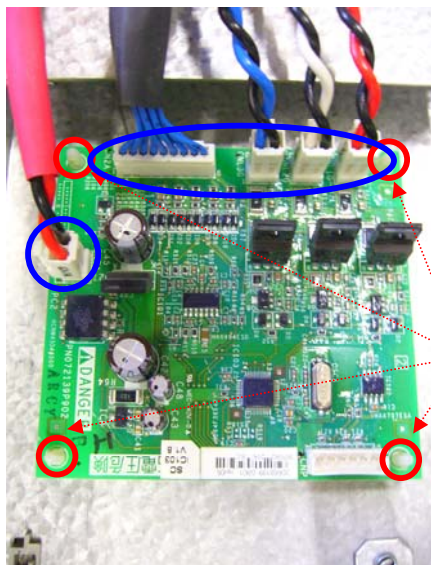
Wire Kit: VZ3N1323



Remove the LED

The LED can be removed.

Soft Charge board: VX5A1300



Disconnect the 5 wires, *left to the right*:

Red CN7A-> CN7 on Power Board

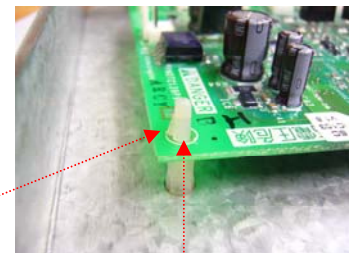
Black CN2A-> CN2 on Power Board

Blue/Black CNL3G -> Gate rectifier 3 on I3

White/Black CNL2G -> Gate rectifier 2 on L2

Red/Black CNL1G -> Gate rectifier 1 on L1

Push on the 4 plastic supports to remove the board.



Push on the side.

The Soft charge Board can be removed.



Size	Torque
M3x8	clips

Assembling Kit: VY1A1303



Remove the 4 screws.

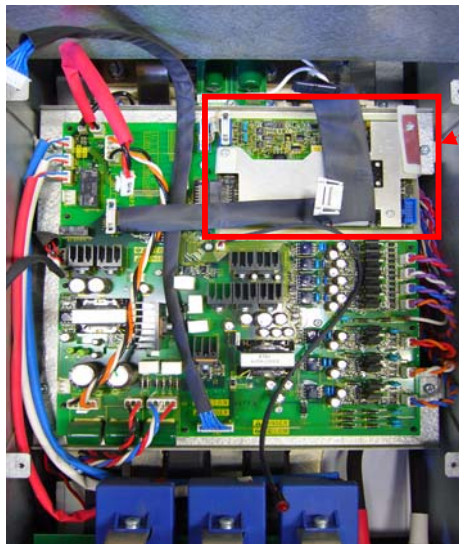
The Assembling kit can be removed.



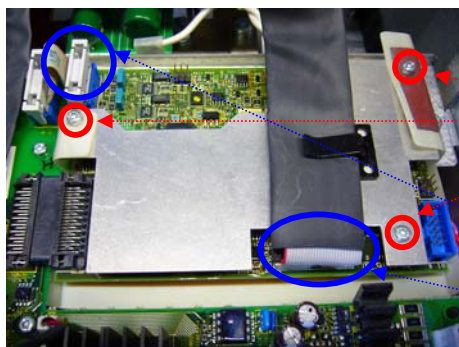
Size	Torque
M4x10	1.2Nm

Power Board: VX5A1HD9011

Part1 motor control board

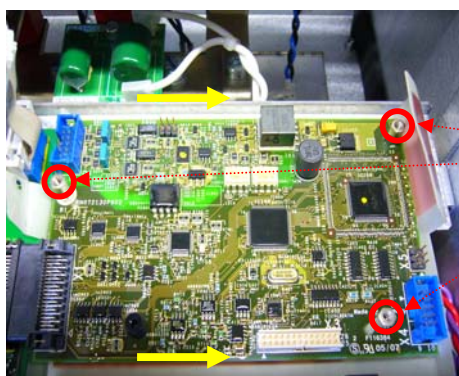


Motor control board



Remove the 3 screws. (P1)

Disconnect the 2 wires, *left to the right*:
X2 -> CNX2 on Power Board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs (P2).

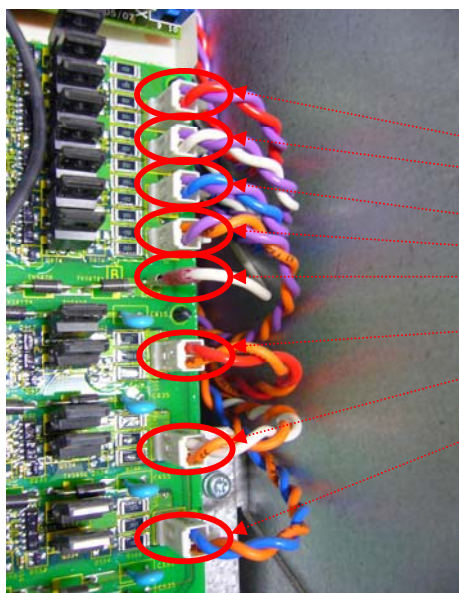
The motor control board can be removed.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

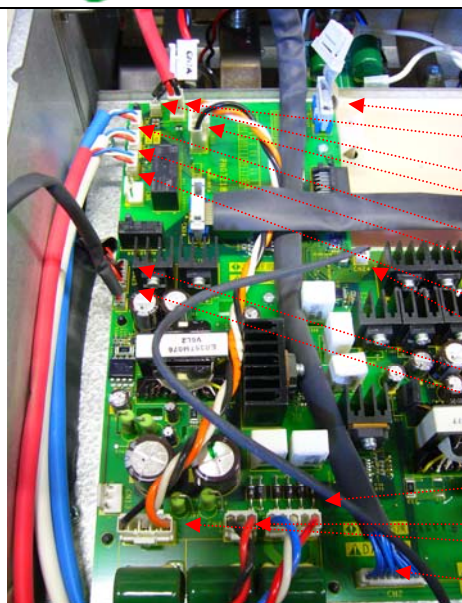
Power Board: VX5A1HD9011

Part2 power board



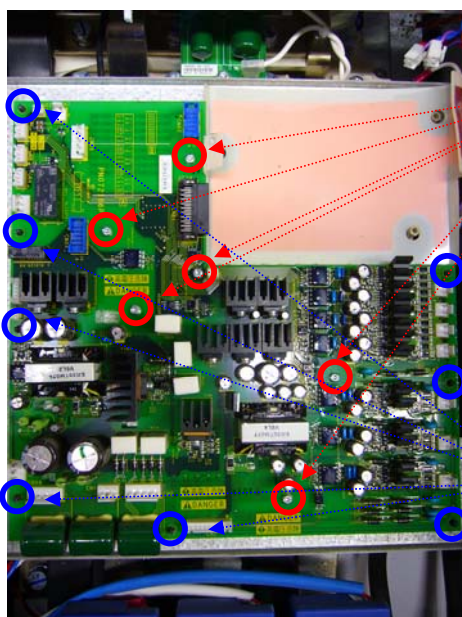
Disconnect the 8 wires, *top to the down*:

CNX	Red/Purple	-> Gate module IGBT
CNY	White/Purple	-> Gate module IGBT
CNZ	Blue/Purple	-> Gate module IGBT
CNPB	Orange/Purple	-> Gate module brake IGBT
TAB1	White	-> Connector 1 of Brake IGBT
CNU	Red/Orange	-> Gate module IGBT
CNV	White/Orange	-> Gate module IGBT
CNW	Blue/Orange	-> Gate module IGBT



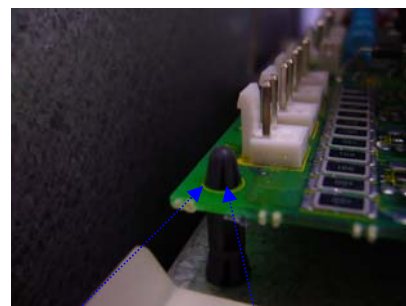
Disconnect the 8 wires, *top to the down*:

- CNX2 Grey -> X2 on motor control board
- CN7 Red -> CN7A on soft charge Board
- CN22 Black -> temperature sensor
- CNX5 Black/White/Orange -> CNX6 on Power board
- CNC Blue -> current sensor W
- CNB White -> current sensor V
- CNA Red -> current sensor U
- CN24 Black -> Power Led
- CN25 Black -> external Fan
- CN12 Black -> internal Fan
- CN5 Blue/White/Red -> Alimentation on input bus bar
- CN6 Black/Red -> Capacitor
- CNX6 Black/White/Orange -> CNX5 on Power board
- CN2 Black -> CN2A on Soft charge Board



Remove the 6 screws. (P1)

Push on the 8 plastic supports to remove the board. (P2)



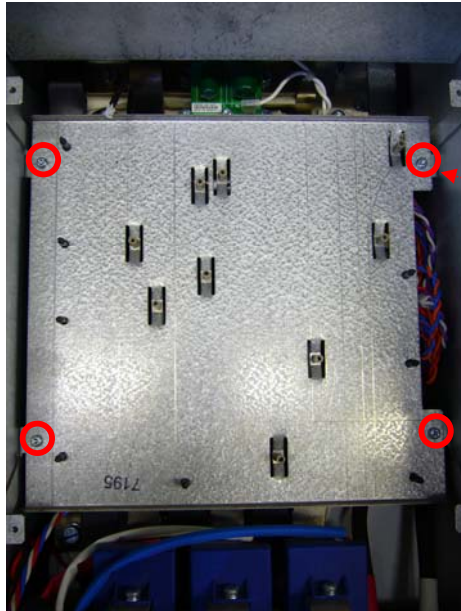
Push on the side.

The Power board can be removed.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x8	Clips

Assembling Kit: VY1A1303



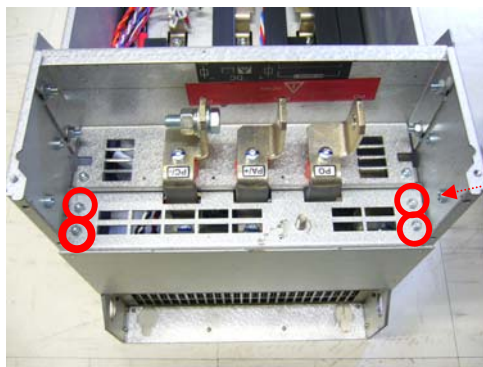
Remove the 4 screws

The Assembling kit can be removed.

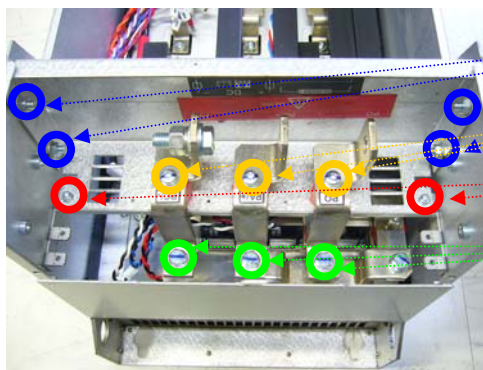


Size	Torque
M4x10	1.2Nm

Kit Bus Bar AC: VZ3N1325



To remove steel, unscrews the 4 screws (P2).



Remove the 4 screws (P1).

Remove the 3 screws (P3).

Remove the 2 screws (P2).

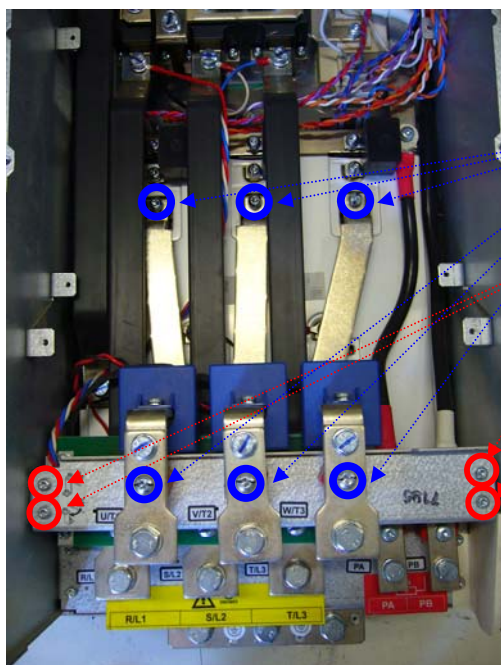
Remove the 3 screws (P4).



Mark	Size	Torque
P1	M6	4.4Nm
P2	M4x10	1.2Nm
P3	M6x14	4.4Nm
P4	M8x20	10.8Nm

The Bus Bar kit can be removed.

Kit Bus Bar AC: VZ3N1325



Remove the 6 screws (P1).

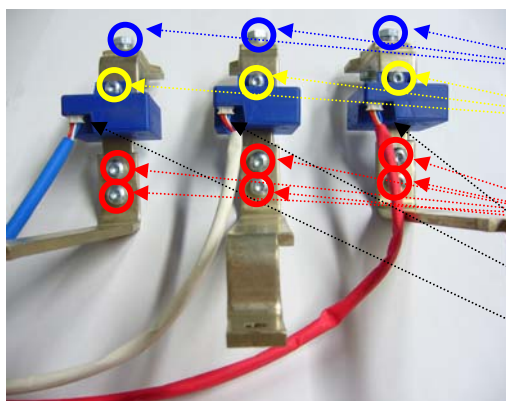
Remove the 4 screws (P2).



Mark	Size	Torque
P1	M6x14	4.4Nm
P2	M4x10	1.2Nm

The Bus Bar kit can be removed.

Motor current sensor: VY1A1105



Remove the 3 screws (P1).

Remove the 3 screws (P2).

Remove the 6 screws (P3).

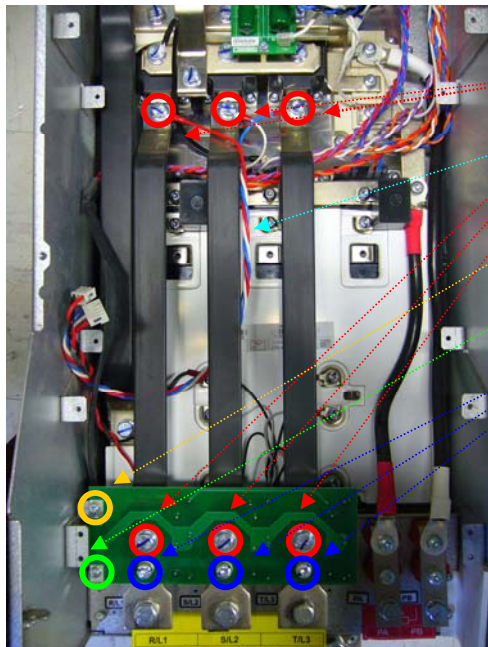
Disconnect the 3 wires, *Right to the left*
Red-> CNA on power board
White-> CNB on power board
Blue-> CNC on power board



Mark	Size	Torque
P1	M8x20	10.8Nm
P2	M3x12	0.8Nm
P3	M6x14	4.4Nm

The motor current sensor can be removed.

RFI Filter Board: VX4A1116 Kit Bus Bar AC: VZ3N1325



Remove the 6 screws (P1).

And disconnect cable Red on L1 White on L2
Blue on L3 CN5-> CN5 on power board

Remove the 1 screw (P2).

Remove the 1 screw (P3).

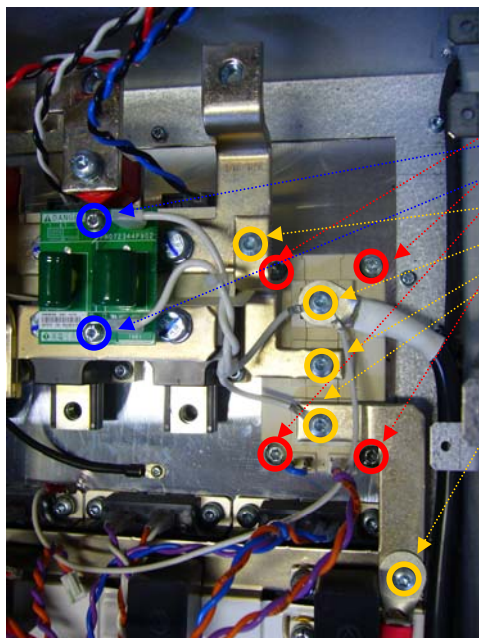
Remove the 3 screws (P4).



Mark	Size	Torque
P1	M8x20	10.8Nm
P3	M4x10	1.2Nm
P4	M6x14	4.4Nm
P2	M4x10	1.2Nm

The RFI filter Board can be removed.

Braking Module IGBT: VZ3TM1400M1271



Remove the 4 screws (P1).

Remove the 2 screws (P2).

Remove the 5 screws (P3).

Disconnect 4 wires, top to the down

White/Black wire -> PB connector
 White wire -> TAB1 on power board
 Orange/Purple wires -> CNPB on power board
 Red/Black wire -> PA connector

Warning: After change the brake IGBT, don't forget to connect Capacitor card between 1&3 of IGBT.
 After changing, **be careful** to set up direction.
Don't forget applying the grease.



Mark	Size	Torque
P1	M6x18	3.0Nm
P2	M4x10	1.2Nm
P3	M6x14	1.2Nm

The Braking module IGBT can be removed.

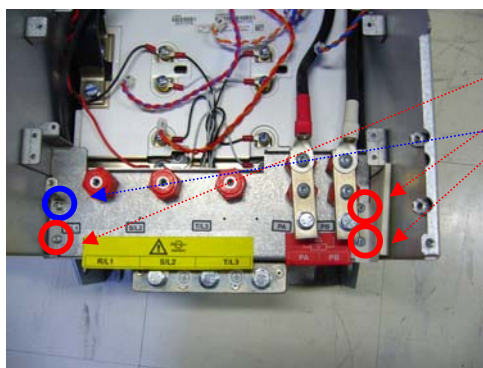
Braking Module IGBT: VZ3TM1400M1271



Be careful, after changing respect the colour line for the wires.

From left to right
[Shunt blue 4&5] – [Purple 7 – Orange 6]

Kit Bus Bar AC: VZ3N1325



Remove the 4 screws (P1)

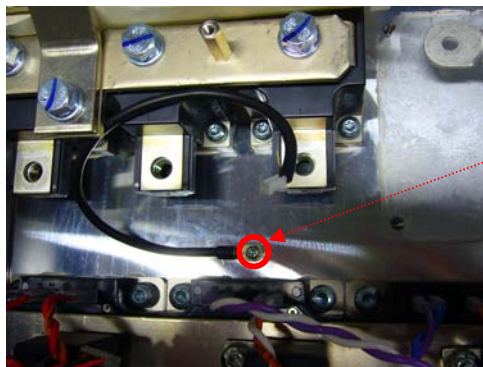
Remove the 1 screw (P2).



Mark	Size	Torque
P1	M4x10	1.2Nm
P2	M4x10	1.2Nm

The Bus Bar kit can be removed.

Thermal sensor: VZ3G1102



Remove the 1 screw.

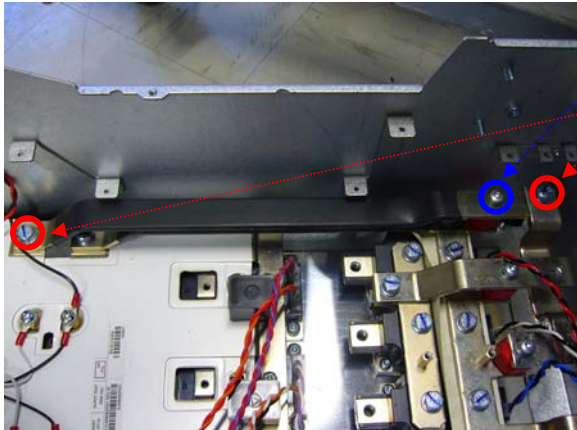
Disconnect 1 wire
Black wire-> CN22 on power board



Size	Torque
M3x6	0.8Nm

The Thermal sensor can be removed.

DC Bus Bar KIT: VZ3N1327



Remove the 1 screw (P1).

Remove the 2 screws (P2).

The first the bar can be removed.



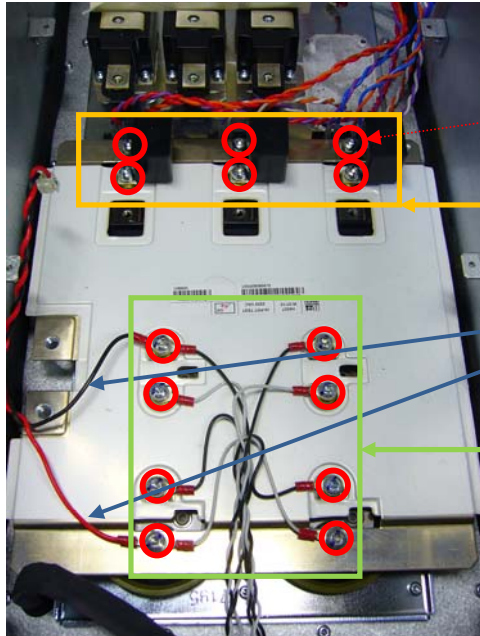
Remove the 7 screws (P2).

The DC Bus Bar KIT can be removed.



Mark	Size	Torque
P1	M6x14	4.4Nm
P2	M8x20	10.8Nm

DC Bus Bar KIT: VZ3N1327



Remove the 14 screws

Zone 2

Disconnect 1wire:
Red/Black wire-> CN6 on power board

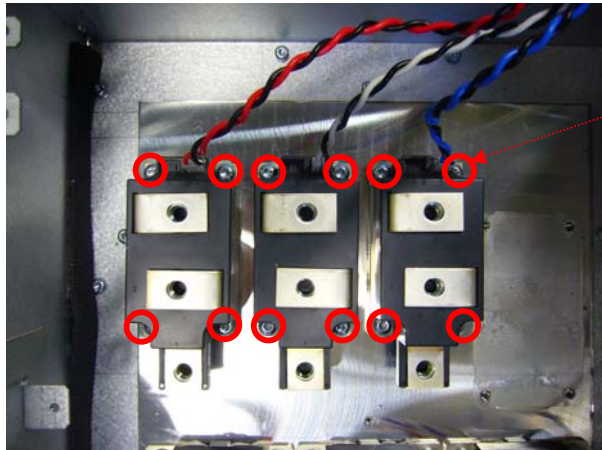
Zone 1

The DC Bus Bar KIT can be removed.



Size	Torque
M6x16	3.3Nm zone 1
	4.4Nm zone 2

Rectifier: VZ3TD1250M1671



Remove the 12 screws

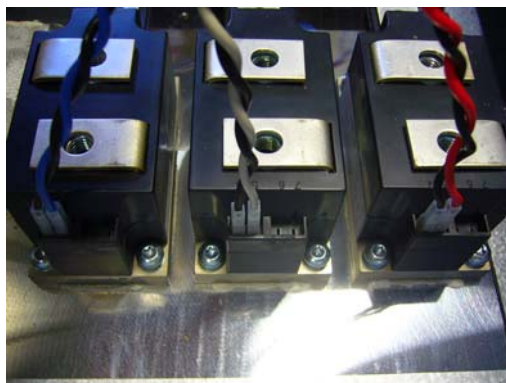
After changing, **be careful** to set up direction.
Don't forget applying the grease.

The Rectifier can be removed.



Size	Torque
M5x18	2.6Nm

Rectifier: VZ3TD1250M1671

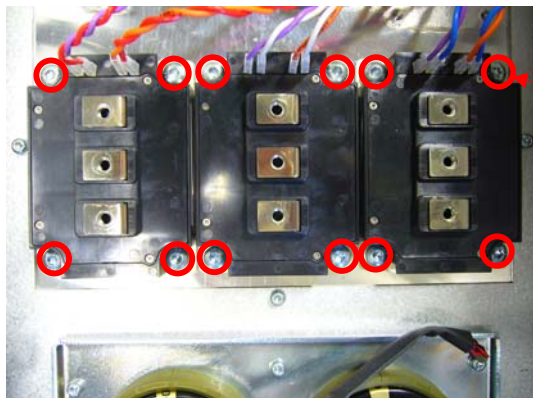


Be careful, after changing respect the colour line for the wires.

From left to right
[Black - **Blue**]- [Black- **White**] - [Black - **Red**]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Module IGBT: VZ3IM1400M1271



Remove the 12 screws.

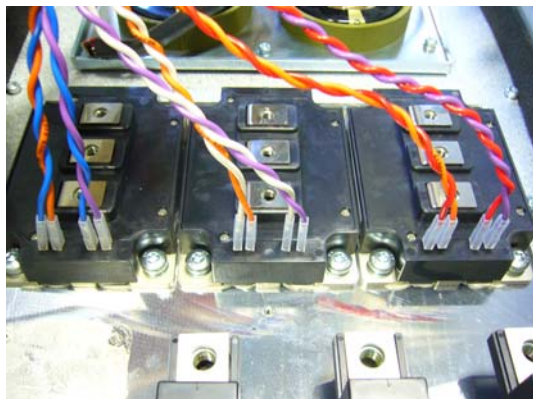
After changing, **be careful** to set up direction.
Don't forget applying the grease.

The Module IGBT can be removed.



Size	Torque
M6x18	4.4Nm

Module IGBT: VZ3IM1400M1271

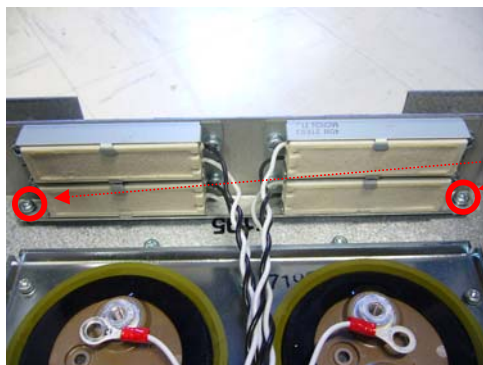


Be careful, after changing respect the colour line for the wires.

From left to right
[Blue – Orange] – [Blue – Purple] – [White- Orange] –
[White- Purple] – [Red – Orange] – [Red – Purple]

It is **not necessary** to change all IGBT but just this one that is damaged.

Discharging resistor: VZ3R24KW040



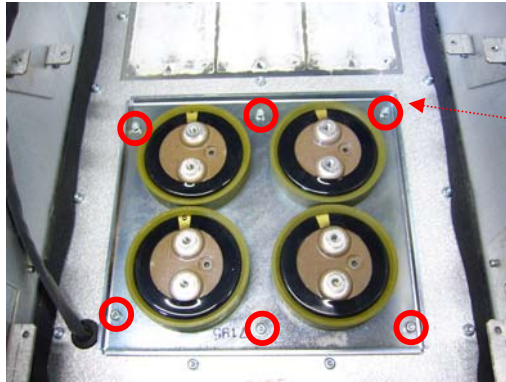
Remove the 2 screws

The Discharging resistor can be removed.



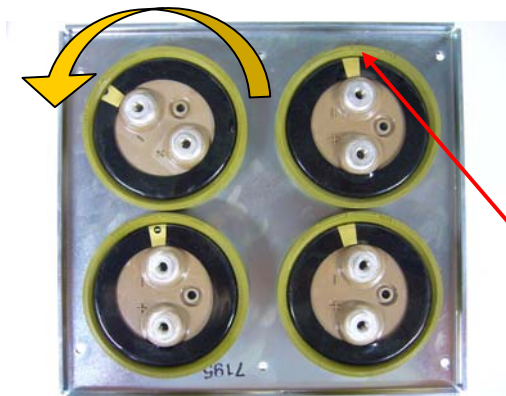
Size	Torque
M4x10	1.2Nm

Lot of 4 Capacitors: VY1ADC1110

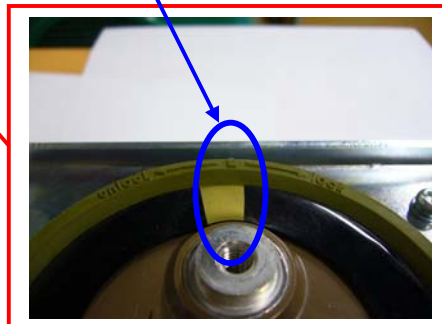


Remove the 6 screws

To remove the capacitor, rotate left it, and pull out it



After changing, **be careful** to set up direction

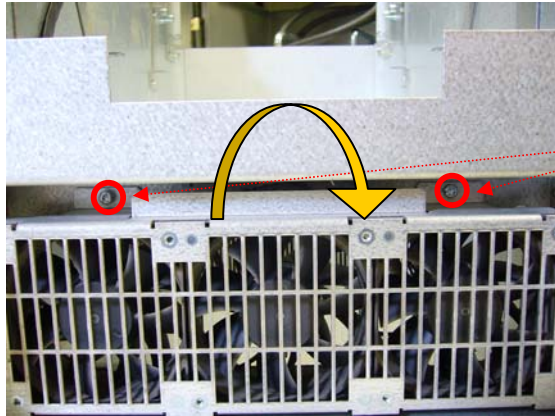


The capacitors can be removed.



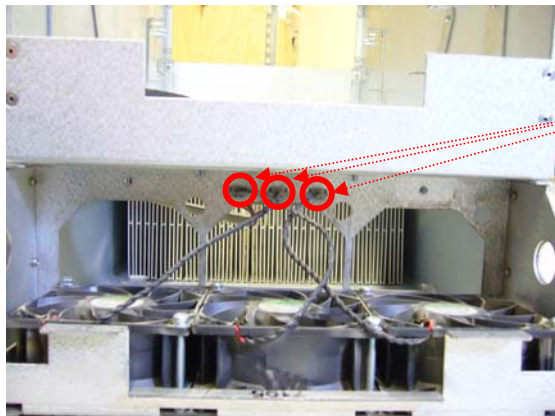
Size	Torque
M4x10	1.2Nm

Fan Kit: VZ3V1215



Remove the 2 screws

Then push fan kit down



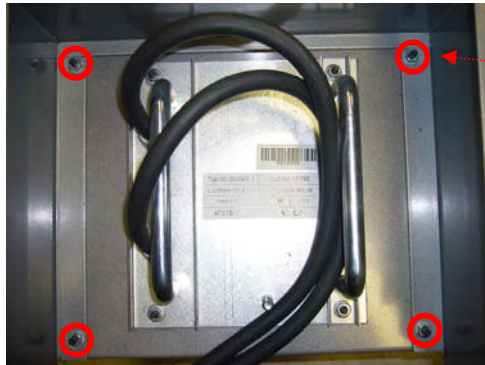
Disconnect the 3 wires.

The fan Kit can be removed.



Size	Torque
M4x10	1.2Nm

DC Choke: VW3A4521



Remove the 4 nuts



Remove the 4 screws



Remove the 4 nuts

The DC choke can be removed.



Size	Torque
M6	5.5Nm
M6x12	5.5Nm



9.21.2 Product Assembling Drawing

No information

9.21.3 Product Cabling Drawing

Refer to following file: [Cabling diagram size 9 200V.tif](#)

Refer to following file: [Cabling diagram size 9 400V.pdf](#)



9.22 ATV71-61 Size 10 (size, refer to 1.2)

9.22.1 Dismantling and reassembling

Frame 10: ATV71HD75M3, ATV71HC11N4, ATV61HD90M3, ATV61HC13N4

ATV71HD75M3	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1216	Fan Power Electronic
VZ3V1214	Internal Fan
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3S1911	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1338	DC bus bar kit for size 10
VZ3N1336	Kit BUS / Bars AC
VZ3N1324	Wires KIT
VZ3IM1600M0671	Module IGBT 600A / 600V
VZ3G1102	Thermal Sensor
VY1ADV1106	Screw kit for size 10
VY1ADC1116	Lots of 4 capacitors DC bus 6800µF / 400V
VY1A1405	Plastic KIT
VY1A1304	Assembling KIT
VY1A1211	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD7590	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX4A71101Y	Control bloc P >=90kW
VX4A1116	RFI Filter Board
VW3A4525	DC Choke
VY1ADV1105	Screws KIT

ATV71HC11N4	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits
VZ3V1214	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1400M1271	Braking Module 400A / 1200V
VZ3TD1285M1671	Rectifier Module (Thyristor / Diode) 285A / 1600V
VZ3S1912	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1328	DC BUS BAR KIT
VZ3N1326	Kit BUS / Bars AC
VZ3N1324	Wires KIT
VZ3IM1400M1271	Module IGBT 400A / 1200V
VZ3G1102	Thermal Sensor
VY1ADV1106	Screws KIT
VY1ADC1111	Lots of 6 capacitors DC bus 6800µF / 400V
VY1A1405	Plastic Parts KIT
VY1A1304	Assembling KIT
VY1A1211	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HC1113	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX4A71101Y	Control bloc P >=90kW
VX4A1116	RFI Filter Board
VW3A4523	DC Choke

ATV61HD90M3	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front Cover 4x7 Digits
VZ3V1216	Fan Power Electronic
VZ3V1214	Internal Fan
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3S1911	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1338	Kit bus bar DC size 10
VZ3N1336	Kit BUS / Bars AC
VZ3N1324	Wires KIT
VZ3IM1600M0671	Module IGBT 600A / 600V
VZ3G1102	Thermal Sensor
VY1ADV1106	Kit visserie
VY1ADC1116	Lots of 4 capacitors DC bus 6800µF / 400V
VY1A1405	Plastic KIT
VY1A1304	Assembling KIT
VY1A1211	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HD7590	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX4A61101Y	Control board P >= 90kW
VX4A1116	RFI Filter Board
VW3A4525	DC Choke

ATV61HC13N4	
Reference	Designation
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits
VZ3V1214	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1400M1271	Braking Module 400A / 1200V
VZ3TD1285M1671	Rectifier Module (Thyristor / Diode) 285A / 1600V
VZ3S1912	Snubber Capacitor KIT
VZ3R24KW040	Discharging Resistor 21k / 40W
VZ3N1328	DC BUS BAR KIT
VZ3N1326	Kit BUS / Bars AC
VZ3N1324	Wires KIT
VZ3IM1400M1271	Module IGBT 400A / 1200V
VZ3G1101	Thermal Sensor
VY1ADV1106	Screws KIT
VY1ADC1111	Lots of 6 capacitors DC bus 6800µF / 400V
VY1A1405	Plastic Parts KIT
VY1A1304	Assembling KIT
VY1A1211	Front Cover incl.I/O Terminal Cover
VY1A1105	Motor Current Sensor
VX5A1HC1113	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX4A1116	RFI Filter Board
VW3A4523	DC Choke
VZ3G1102	Thermal Sensor
VX4A61101Y	Control board P >= 90kW

Front cover with I/O terminal cover: VY1A11211



Remove 7 screws

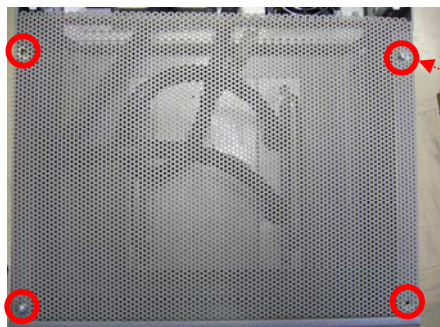


The Front cover can be removed.



Size	Torque
M6x10	4.4Nm

DC Choke Front cover



Remove 4 screws

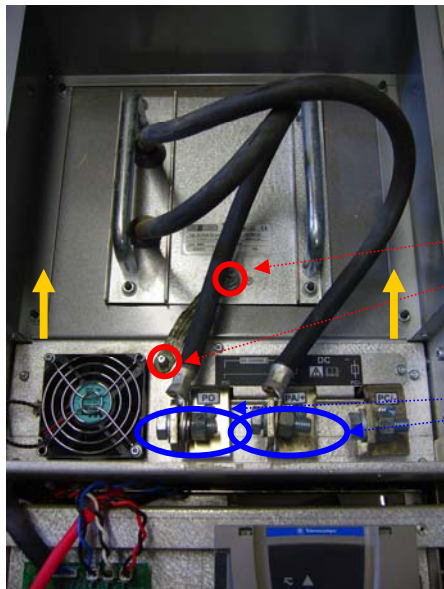


The DC Choke front cover can be removed.



Size	Torque
M6x12	4.4Nm

DC Choke connection



Remove 2 nuts (P1).

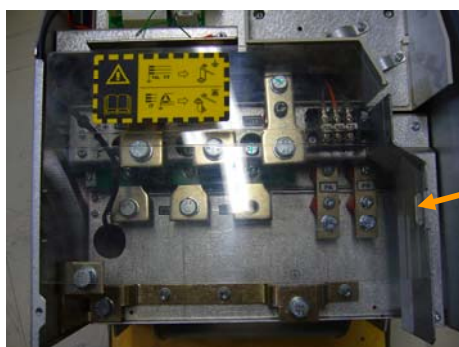
Remove 2 screws (P2).



Mark	Size	Torque
P1	M8	10.8Nm
P2	M12x25	45Nm

The DC Choke can be removed.

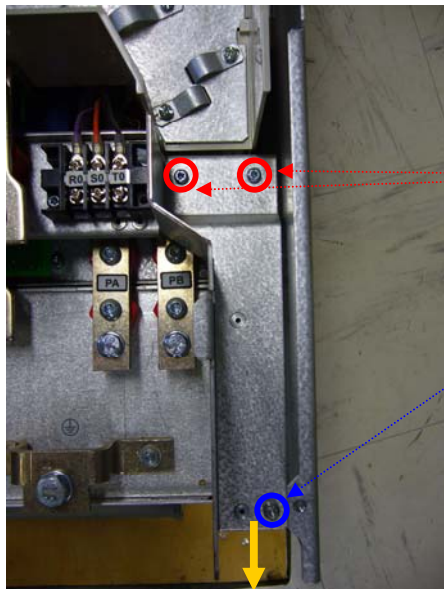
Plastic part KIT: VY1A1405



Push here and disengage plastic part

The plastic part kit can be removed.

Assembling KIT: VY1A1304



Remove 2 screws (S?).

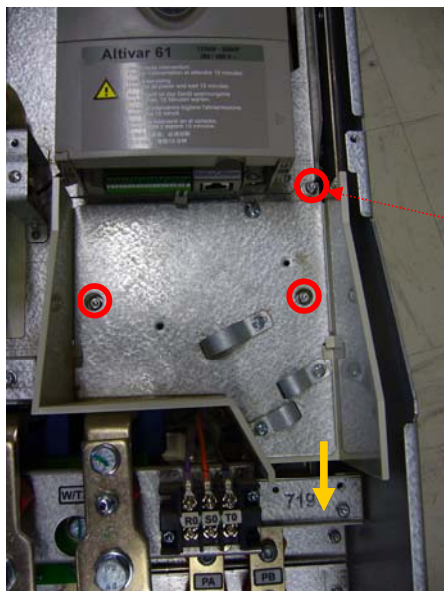
Remove 1 screw (S?).



Mark	Size	Torque
P1	M4x10	1.2Nm
P2	M4x20	0.8Nm

The Part can be removed.

Plastic part KIT: VY1A1405



Remove 3 screws



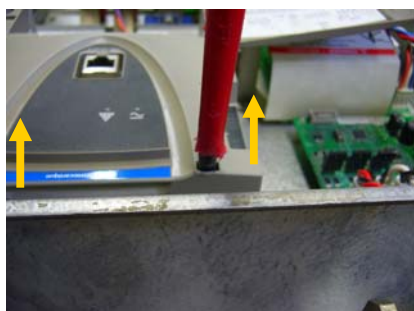
Size	Torque
M4x10	1.2Nm

The Plastic Part kit can be removed.

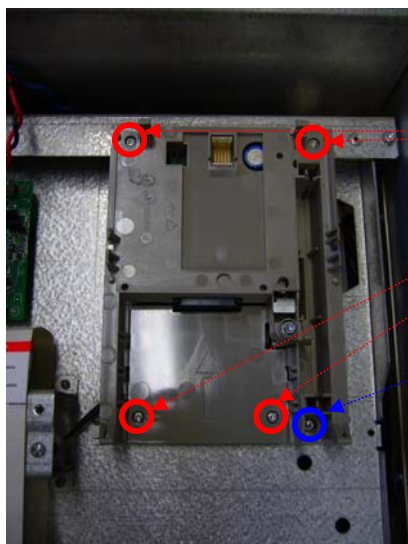
Control bloc: VX4A71101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board

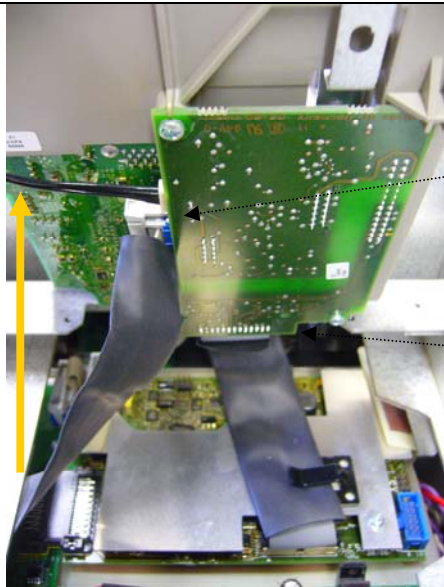


Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 – S8).

Remove 1 screw (S38).



Disconnect the ribbon cable.
X4 from Control bloc" interface Board"->CNX4 on Power Board

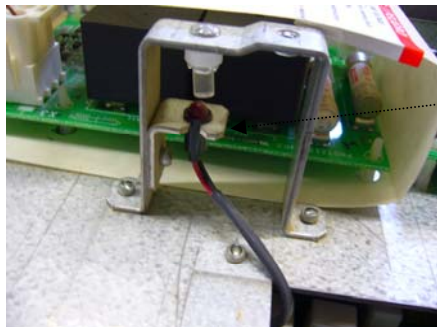
Disconnect the ribbon cable.
X3 from Control bloc "interface Board"->X3 on Motor Control Board



Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5Nm

The Control bloc can be removed.

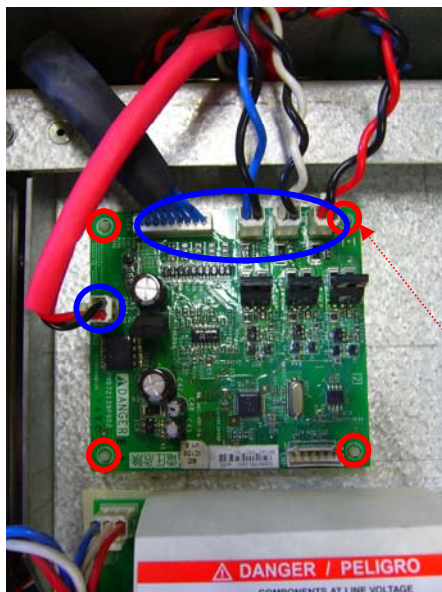
Wire Kit: VZ3N1324



Remove the LED

The LED can be removed.

Soft Charge board: VX5A1300



Disconnect the 5 wires, *left to the right*:

Red CN7A-> CN7 on Power Board

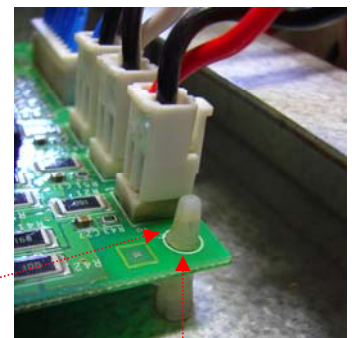
Black CN2A-> CN2 on Power Board

Blue/Black CNL3G ->Gate rectifier 3 on I3

White/Black CNL2G ->Gate rectifier 2 on L2

Red/Black CNL1G -> Gate rectifier 1 on L1

Push on the 4 plastic supports to remove the board.



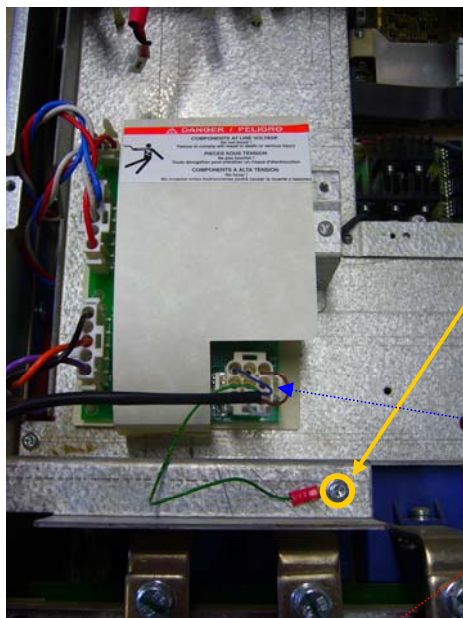
Push on the side.

The Soft charge Board can be removed.



Size	Torque
M3x8	clips

Fan Power Electronic. VZ3V1212

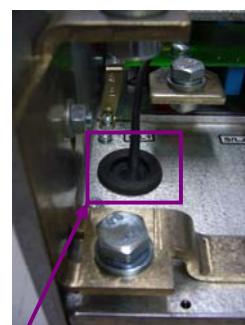


Be careful; don't forget to disconnect the ground wire.

Disconnect 1 connector

Fan turbine 1->X3 on fan control board

Remove 2 nuts



Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

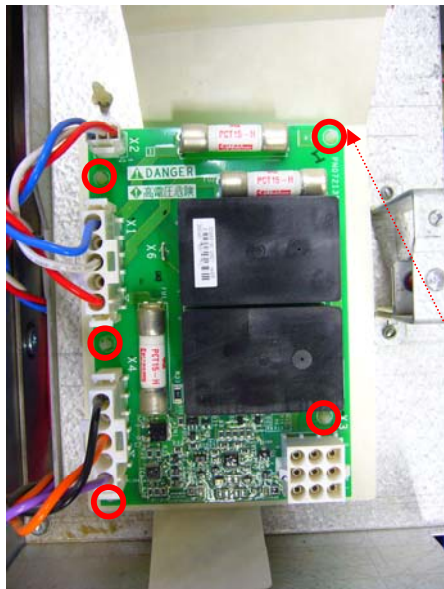


The Turbine can be removed.



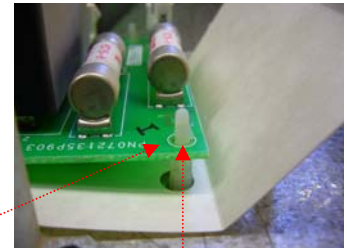
Size	Torque
M6	5.5Nm

Fan control board: VX5A1400



Disconnect the 5 wires, *top to the down*:
 Red/Blue/White X2-> CN13 on Power Board
 Red/White/Blue X1-> Red on L1 White on L2 Blue on L3
 Black/Orange/Purple X4 ->external fan connector TB1 on kit **VY1A1304**
 X3 -> turbine alimentation

Push on the 5 plastic supports
to remove the board.



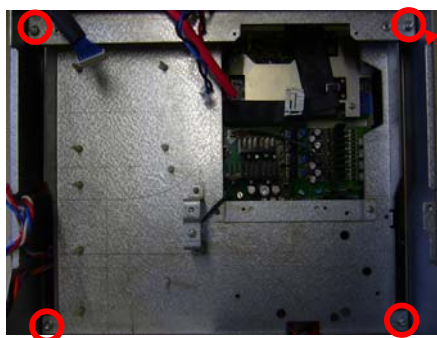
Push on the side.

The Soft charge Board can be removed.



Size	Torque
M3x8	clips

Assembling KIT: VY1A1304



Remove 4 screws

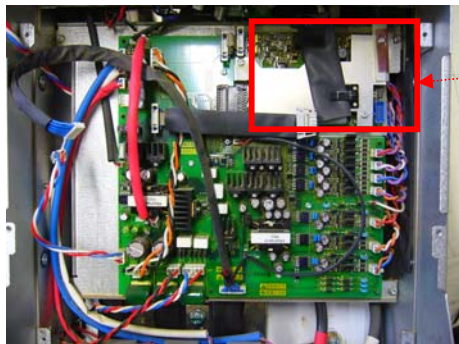
The part can be removed.



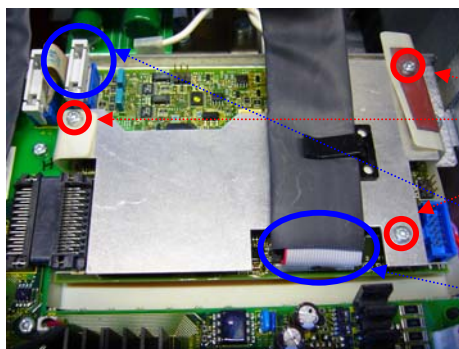
Size	Torque
M4x10	1.2Nm

Power Board: VX5A1HC1113

Part1 motor control board

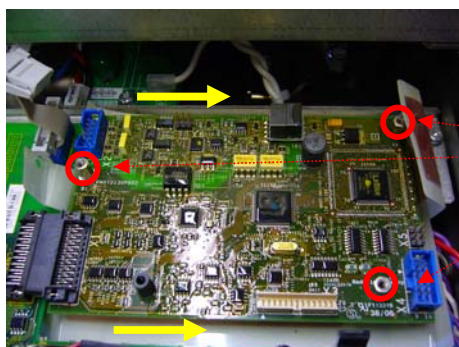


Motor control board



Remove the 3 screws. (P1)

Disconnect the 2 wires, *left to the right*:
X2 -> CNX2 on Power Board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs.(P2)

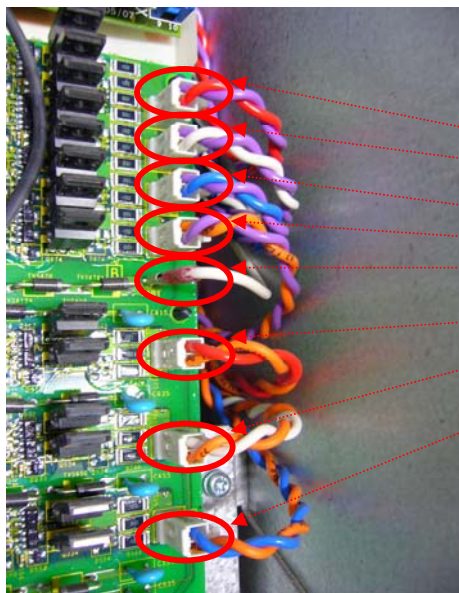
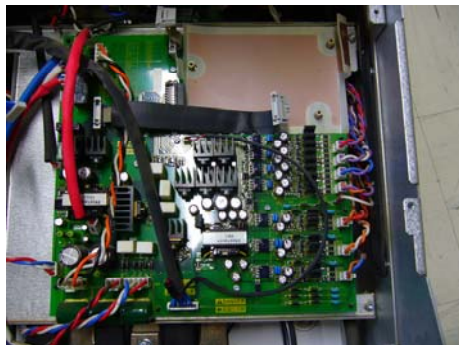


Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

The motor control board can be removed.

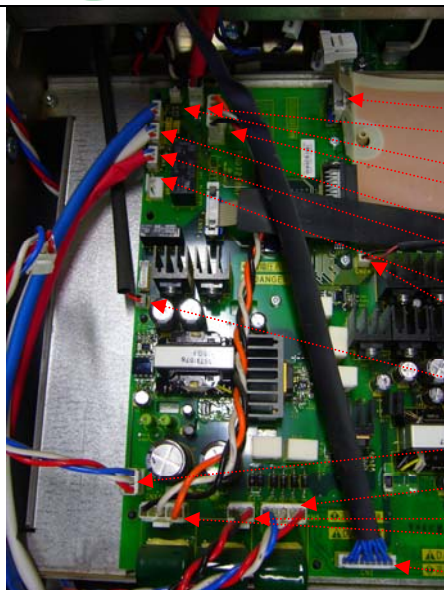
Power Board: VX5A1HC1113

Part2 power board



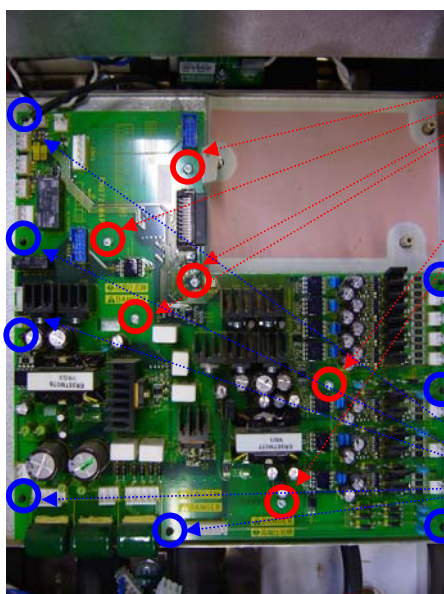
Disconnect the 8 wires, *top to the down*:

CNX	Red/Purple	-> Gate module IGBT
CNY	White/Purple	-> Gate module IGBT
CNZ	Blue/Purple	-> Gate module IGBT
CNPB	Orange/Purple	-> Gate module brake IGBT
TAB1	White	-> Connector 1 of Brake IGBT
CNU	Red/Orange	-> Gate module IGBT
CNV	White/Orange	-> Gate module IGBT
CNW	Blue/Orange	-> Gate module IGBT



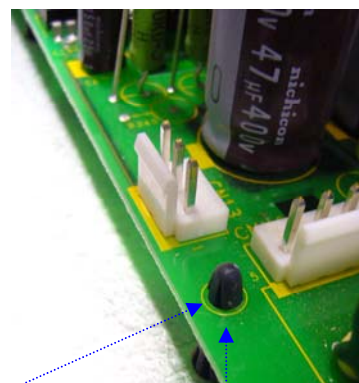
Disconnect the 8 wires, *top to the down*:

- CNX2 Grey -> X2 on motor control board
- CN7 Red -> CN7A on soft charge Board
- CN22 Black -> temperature sensor
- CNX5 Black/White/Orange -> CNX6 on Power board
- CNC Blue -> current sensor W
- CNB White -> current sensor V
- CNA Red -> current sensor U
- CN24 Black -> Power Led
- CN12 Black -> internal Fan
- CN13 Red/Blue/White -> X2 on Fan control Board
- CN5 Blue/White/Red -> Alimentation on input bus bar
- CN6 Black/Red -> Capacitor
- CNX6 Black/White/Orange -> CNX5 on Power board
- CN2 Black -> CN2A on Soft charge Board



Remove the 6 screws. (P1)

Push on the 8 plastic supports
to remove the board. (P2)



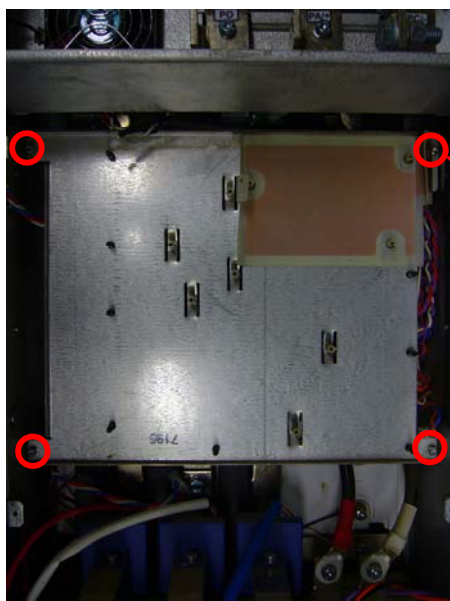
Push on the side.

The Power board can be removed.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x8	clips

Assembling KIT: VY1A1304



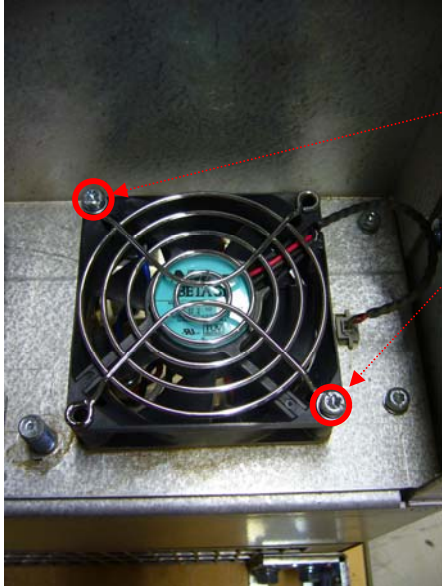
Remove the 4 screws

The part can be removed.



Size	Torque
M4x10	1.2Nm

Internal Fan: VZ3V1214



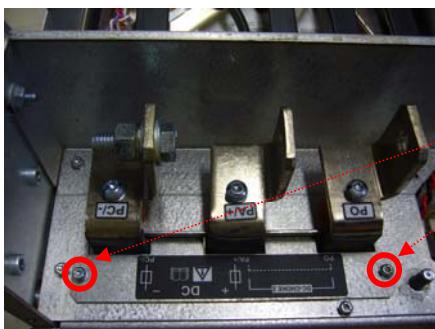
Remove the 2 screws

The internal fan can be removed.



Size	Torque
M4x35	1.2Nm

Assembling kit: VY1A1304 Part 1



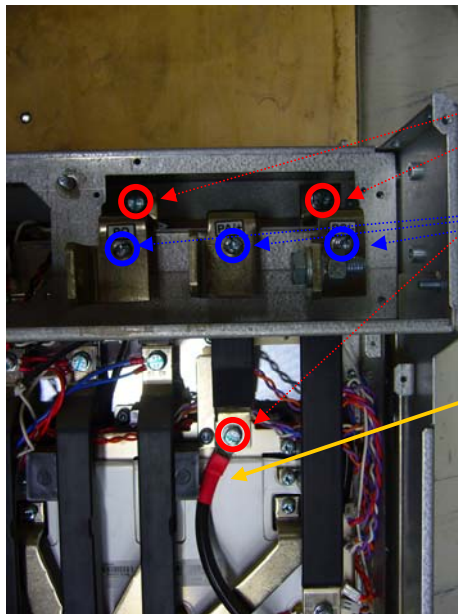
Remove the 2 screws

The part can be removed.



Size	Torque
M4x10	1.2Nm

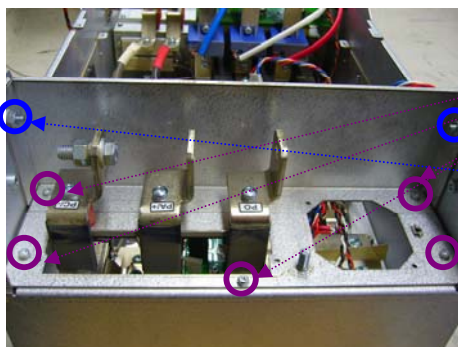
Assembling kit: VY1A1304 & bars kit: VZ3N1328
Part 2



Remove the 3 screws (P2).

Remove the 3 screws (P1).

Disconnect 1 wire
Red/Black wire -> PA connector



Remove the 5 screws (P4).

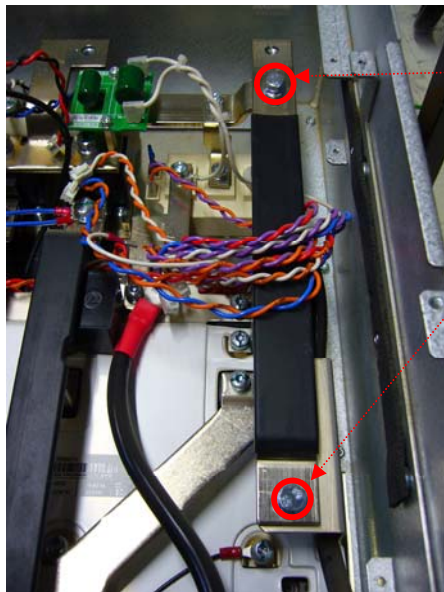
Remove the 2 nuts (P3).

The Assembly kit can be removed.



Mark	Size	Torque
P1	M6x14	4.4Nm
P2	M8x20	10.8Nm
P3	M6	4.4Nm
P4	M4x10	1.2Nm

DC Bus bars kit: VZ3N1328



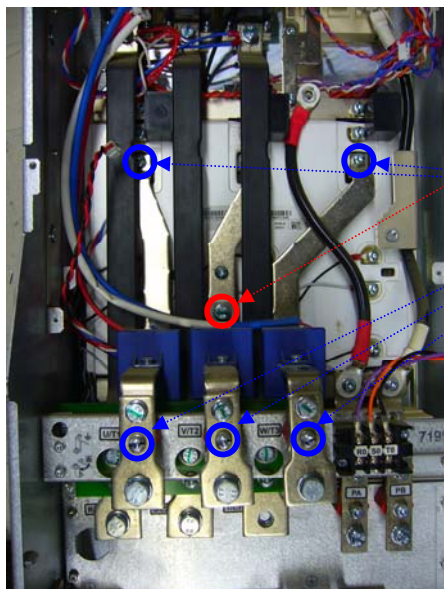
Remove the 2 screws

The DC Bus bar kit can be removed.



Size	Torque
M8x20	10.8Nm

AC Bus bars kit: VZ3N1326



Remove the 1 screw (P1).

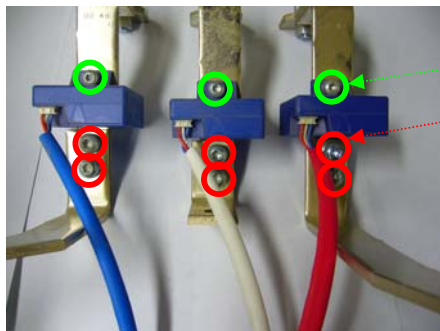
Remove the 5 screws (P2).

The AC Bus bar kit can be removed.



Mark	Size	Torque
P1	M6x12	4.4Nm
P2	M6x14	4.4Nm

Motor current sensor: VY1A1105



Remove 3 screws (P1)

Remove the 6 screw (P2).

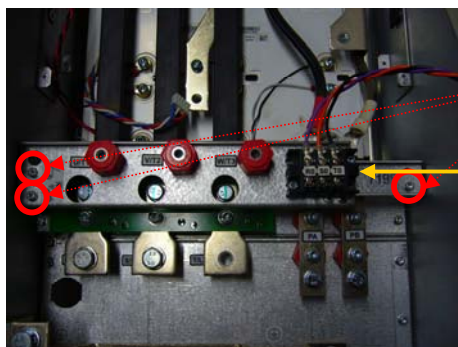
Disconnect the 3 wires, *Right to the left*
Red-> CNA on power board
White-> CNB on power board
Blue-> CNC on power board



Mark	Size	Torque
P1	M3x12	0.8Nm
P2	M6x14	4.4Nm

The Motor current sensor can be removed.

Assembling kit: VY1A1304



Remove the 3 screws

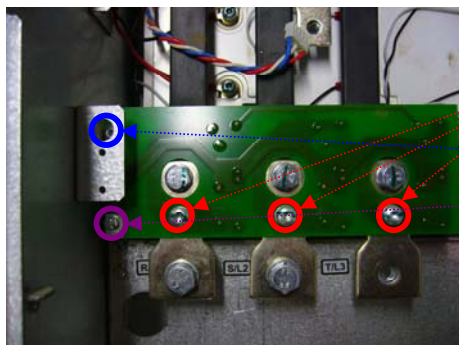
Connector TB1 wire Purple/Orange/Black -> X4 on fan control board



Size	Torque
M4x10	1.2Nm

The part can be removed.

RFI Filter Board: VX4A1116



Remove the 3 screws (P1).

Remove the 1 screw (P2)

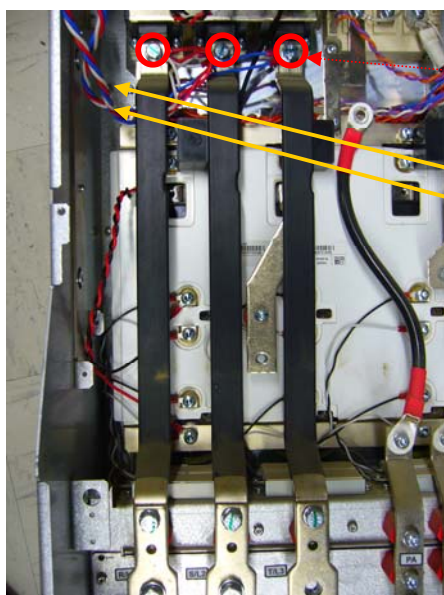
Remove the 1 screw (P3)



Mark	Size	Torque
P1	M6x14	4.4Nm
P2	M4x10	1.2Nm
P3	M4x10	1.2Nm

The RFI filter board can be removed.

AC Bus bars kit: VZ3N1326



Remove the 3 screws

Disconnect 2 wires:

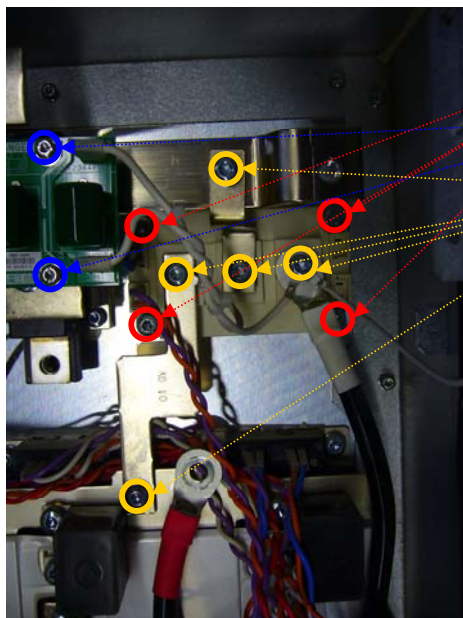
Red L1/white L2/Blue L3->X1 on Fan control board
Red L1/white L2/Blue L3->CN5 on Power board

The AC Bus bar kit can be removed.



Size	Torque
M8x20	10.8Nm

Brake IGBT: VZ3TM1400M1271



Remove the 4 screws (P1).

Remove the 2 screws (P2).

Remove the 5 screws (P3).

Disconnect 4 wires, Right to the Left

White/Black wire -> PB connector

White wire -> TAB1 on power board

Orange/Purple wires -> CNPB on power board

Warning: After change the brake IGBT, don't forget to connect Capacitor card between 1&3 of IGBT.

After changing, **be careful** to set up direction.

Don't forget applying the grease.



Mark	Size	Torque
P1	M6x18	3.0Nm
P2	M4x10	1.2Nm
P3	M6x14	1.2Nm

The Brake IGBT can be removed.

Brake IGBT: VZ3TM1400M1271

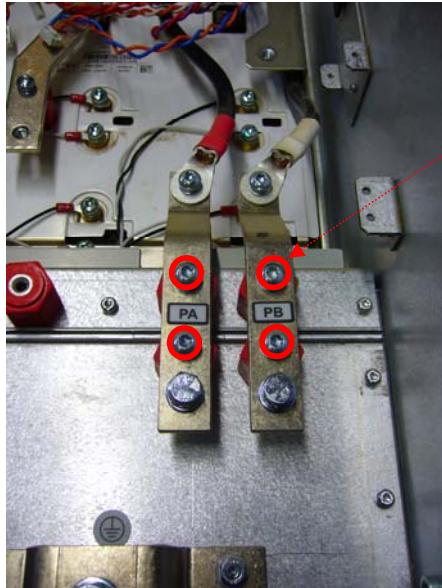


Be careful, after changing respect the colour line for the wires.

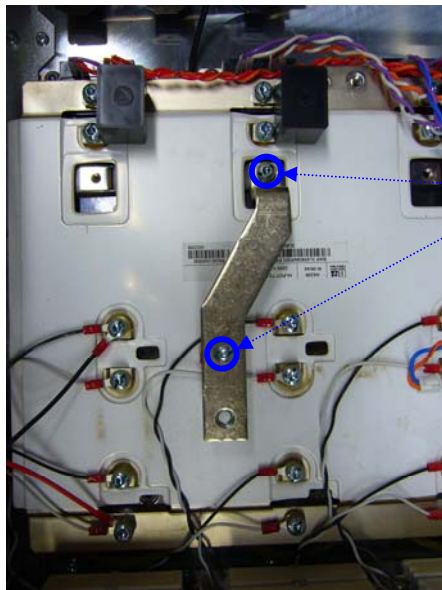
From left to right

[Shunt blue 4&5] – [Purple 7 – Orange 6]

AC Bus bars kit: VZ3N1326



Remove the 4 screws (P1).



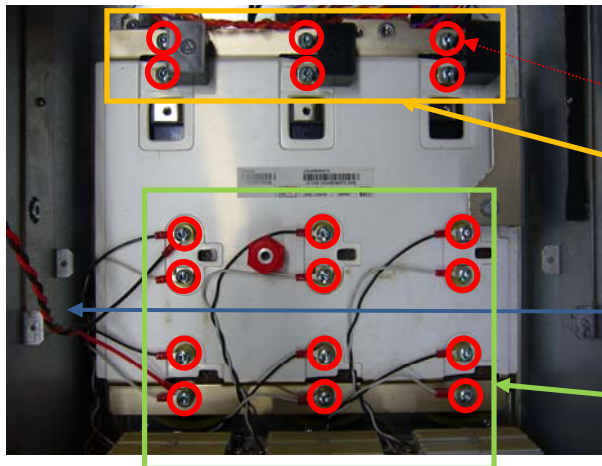
Remove the 2 screws (P2).



Mark	Size	Torque
P1	M6x14	4.4Nm
P2	M6x16	4.4Nm

The AC Bus bar kit can be removed.

DC Bus Bar: VZ3N1328



Remove the 18 screws

Zone 2

Disconnect 1 wire:
Red/Black wire -> CN6 on power board

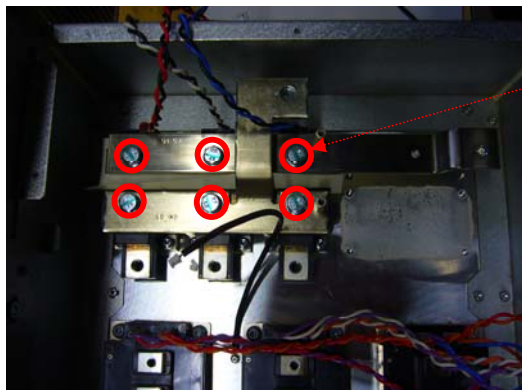
Zone 1

The DC Bus Bar can be removed.



Size	Torque
M6x16	3.3Nm zone 1 4.4Nm zone 2

Rectifier DC Bus Bar: VZ3N1328



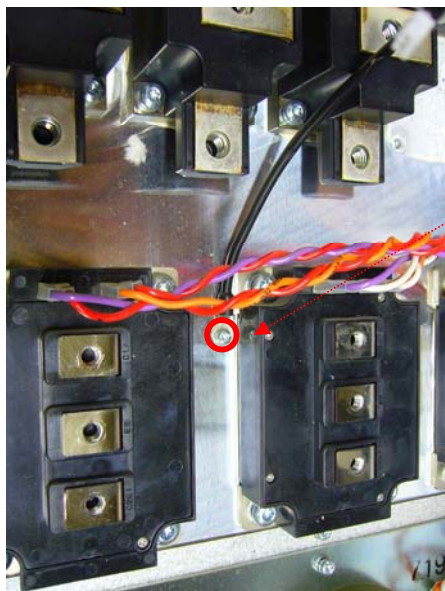
Remove the 6 screws

The DC Bus Bar can be removed



Size	Torque
M8x20	10.8Nm

Thermal sensor: VZ3G1101



Remove the 1 screw

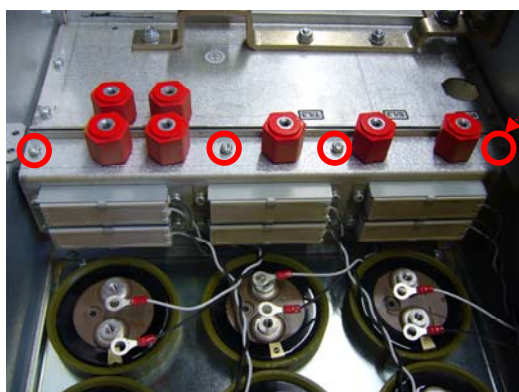
Disconnect 1 wire
Black wire-> CN22 on power board

The Thermal sensor can be removed.



Size	Torque
M3x6	0.8Nm

Discharging Resistor: VZ3R24KW040



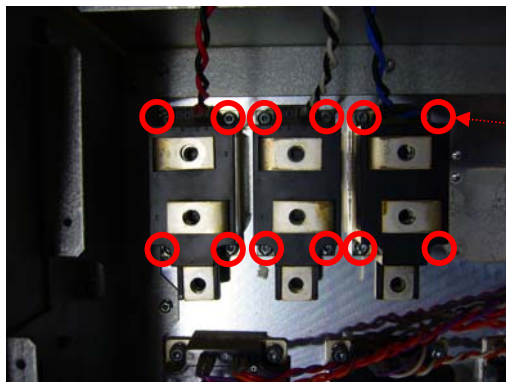
Remove the 4 screws

The discharging resistors can be removed.



Size	Torque
M4x10	1.2Nm

Rectifier module: VZ3TD1285M1671



Remove the 12 screws

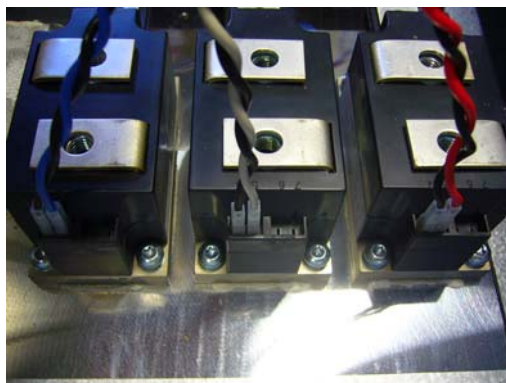
After changing, **be careful** to set up direction.
Don't forget applying the grease.



Size	Torque
M5x18	2.6Nm

The Rectifier can be removed.

rectifier: VZ3TD1285M1671

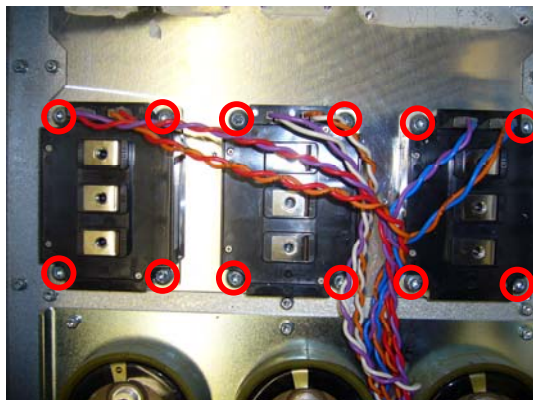


Be careful, after changing respect the colour line for the wires.

From left to right
[Black - Blue]- [Black- White] - [Black - Red]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Module IGBT: VZ3IM1400M1271



Remove the 12 screws

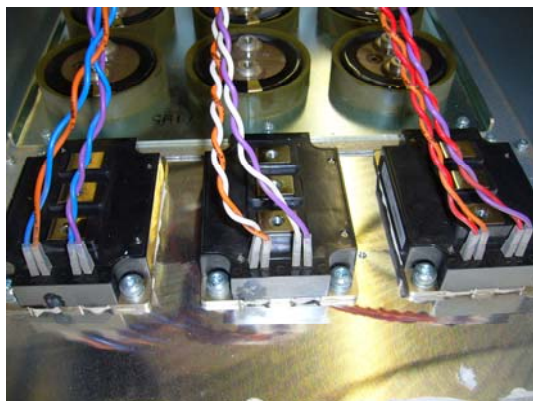
After changing, **be careful** to set up direction.
Don't forget applying the grease.

The IGBT module can be removed.



Size	Torque
M6x18	4.4Nm

Module IGBT: VZ3IM1400M1271



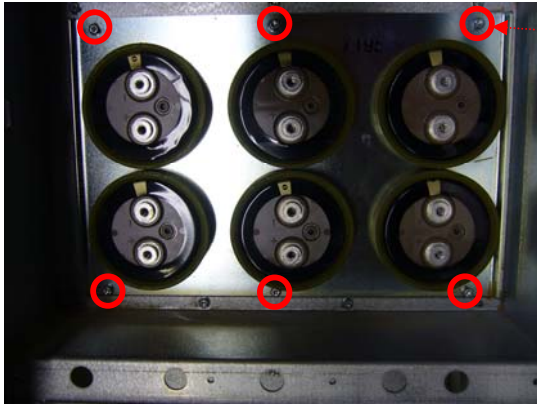
Be careful, after changing respect the colour line for the wires.

From left to right

[Blue – Orange] – [Blue – Purple] – [White- Orange] –
[White- Purple] – [Red – Orange] – [Red – Purple]

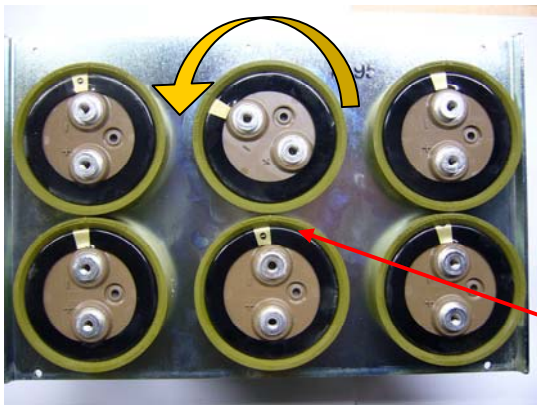
It is **not necessary** to change all IGBT but just this one that is damaged.

Lot of Capacitors: VY1ADC1111

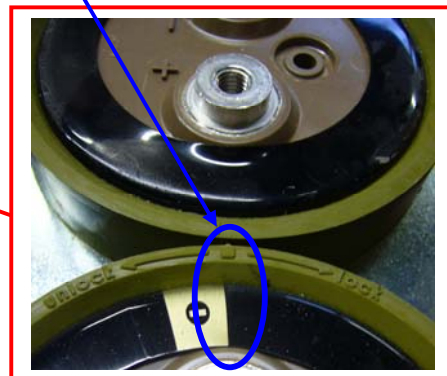


Remove the 6 screws.

To remove the capacitor, rotate left it, and pull out it



After changing, **be careful** to set up direction



The capacitor can be removed.



Size	Torque
M4x10	1.2Nm

DC SHOKE: VW3A4523



Remove the 4 nuts



Remove the 4 screws



Remove the 4 nuts

The DC Choke can be removed.



Size	Torque
M6	5.5Nm
M6x12	5.5Nm



9.22.2 Product Assembling Drawing

No information

9.22.3 Product Cabling Drawing

Refer to following file: [Cabling diagram size 10 200V.tif](#)

Refer to following file: [Cabling diagram size 10 400V.pdf](#)



9.23 ATV61/71 Size 11 (size, refer to 1.2)

9.23.1 Dismantling and reassembling

Size 11: ATV71HC13N4, ATV61HC16N4

ATV71HC13N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TD1285M1671	Rectifier Module (Thyristor / Diode) 285A / 1600V
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1316	Wires KIT
VZ3N1315	AC Bus Bar KIT
VZ3N1314	DC Bus Bar KIT
VZ3IM1600M1271	Module IGBT (600A / 1200V)
VZ3G1104	Thermal Sensor
VZ3F1110	Braking Unit KIT
VY1ADV1107	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200μF / 400V)
VY1A1401	Plastic Parts KIT
VY1A1301	Assembling KIT
VY1A1212	Front Cover with I/O Terminal Cover
VY1A1107	Motor Current Sensor
VX5A71HC13N4	mesuring board 71HC13N4
VX5A1HC1316	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1200	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1113	RFI Filter Board
VW3A4520	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HC16N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1316	Wires KIT
VZ3N1315	AC Bus Bar KIT
VZ3N1314	DC Bus Bar KIT
VZ3IM1600M1271	Module IGBT (600A / 1200V)
VZ3G1104	Thermal Sensor
VZ3F1110	Braking Unit KIT
VY1ADV1107	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200μF / 400V)
VY1A1401	Plastic Parts KIT
VY1A1301	Assembling KIT
VY1A1212	Front Cover with I/O Terminal Cover
VY1A1107	Motor Current Sensor
VX5A71HC13N4	Measuring Board
VX5A1HC1316	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1200	Gate Drive Board
VX4A61101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1113	RFI Filter Board
VW3A4520	DC Choke
VZ3TD1285M1671	Rectifier Module (Thyristor / Diode) 285A / 1600V
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Cover with I/O Terminal Cover: VY1A1212



Remove 9 screws

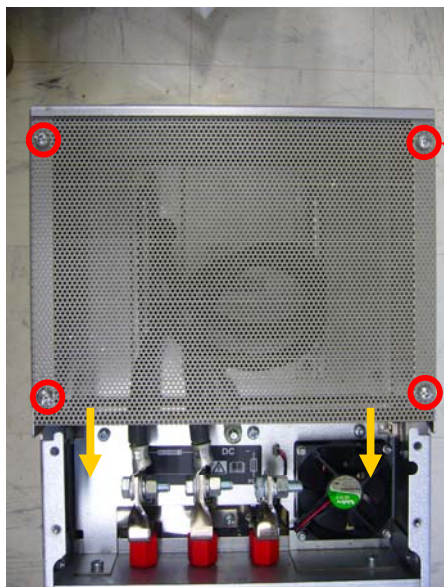


The front cover can be removed.



Size	Torque
M6x12	5.5Nm

Front cover DC Choke: VW3A4520



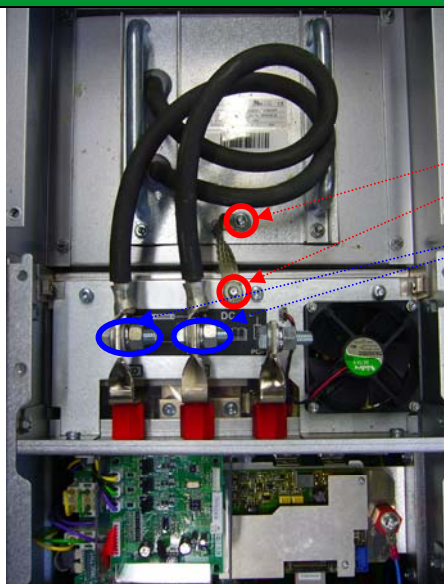
Remove 4 screws (S?).

The DC Choke front cover can be removed.



Size	Torque
M6x12	5.5Nm

DC Choke connection: VW3A4520



Remove 2 screws (P1).

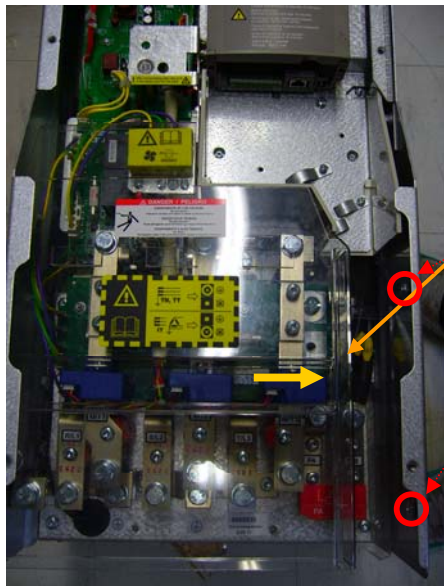
Remove 2 screws (P2).

The DC Choke can be removed.



Mark	Size	Torque
P1	M8	13.5Nm
P2	M12x25	45Nm

Plastic Parts Kit: VY1A1401



Push here and disengage plastic part

Remove 2 screws

The Plastic part KIT can be removed.

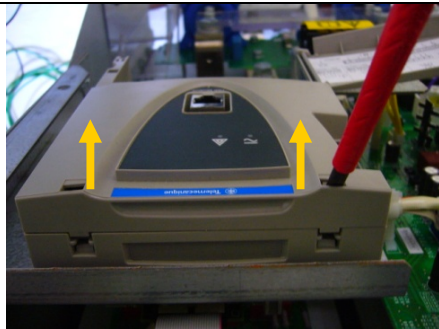


Size	Torque
M6	5.5Nm

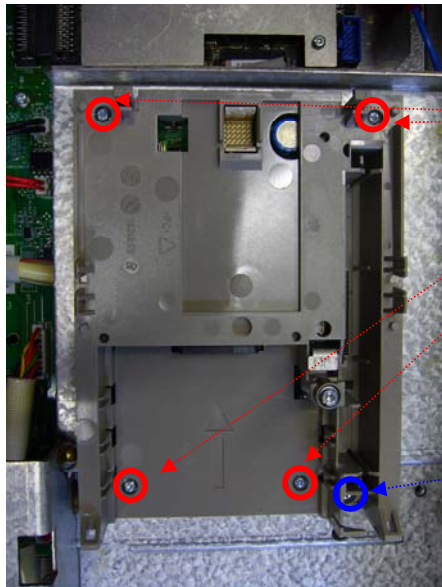
Control bloc: VX4A61101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.



Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 to S8)

Remove 1 screw (S38).

Disconnect the ribbon cable.
X3 from Control bloc "interface Board"->X3 on Motor Control Board

Disconnect the ribbon cable.
X4 from Control bloc" interface Board"->X4 on Measuring Board



The Control bloc can be removed.



Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5NM

Metal Support for Control Bloc



Remove 2 nuts.



Disconnect the wire:

Yellow/Green/Purple on TB1 -> X4 on fan control board

Green -> ground of turbine

Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

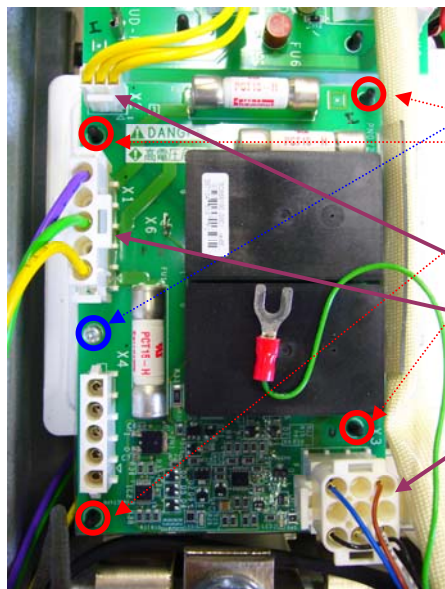
Be careful, don't forget to disconnect the ground wire.

The metal support can be removed.



Size	Torque
M6	5.5Nm

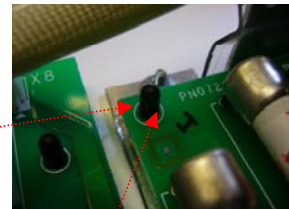
Fan Control Board: VX5A1400



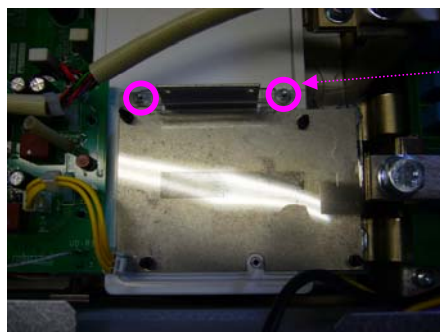
Remove 1 screw

Push on the 4 plastic supports to remove the board.

Disconnect the 3 wires. From right to left:
Yellow X2->X14 on Measuring Board
Yellow/Green/Purple X1-> X1 Yellow X2 Green X3 Purple on filter card
X3-> turbine alimentation



Push on the side.



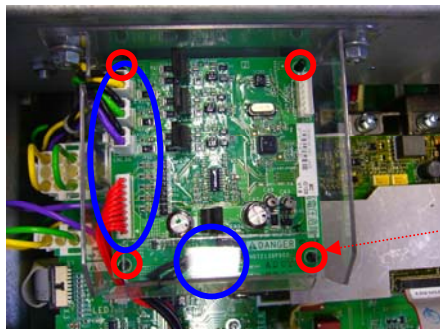
Remove 2 nuts.



Size	Torque
M3x6	0.8Nm
M3x8	clips
M3	0.8Nm

The Fan control board can be removed.

Soft Charge board: VX5A1300



Disconnect the 5 wires, *down to the top*:

Black CN7A-> X31 on Measuring Board

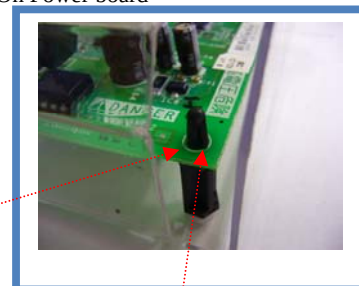
Red CN2A-> X30 on Measuring Board and X30 On Power board

Purple/Black CNL3G ->Gate rectifier 3 on L3

Green/Black CNL2G ->Gate rectifier 2 on L2

Yellow/Black CNL1G ->Gate rectifier 1 on L1

Push on the 4 plastic supports
to remove the board.



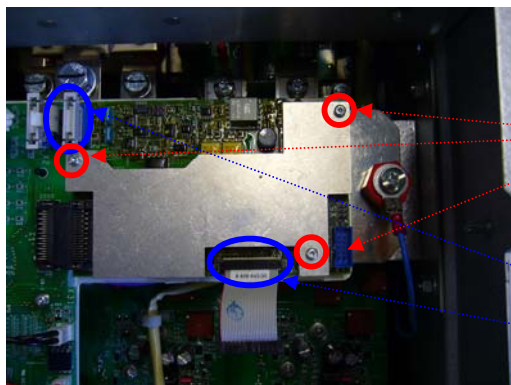
Push on the side.

The Soft charge Board can be removed.



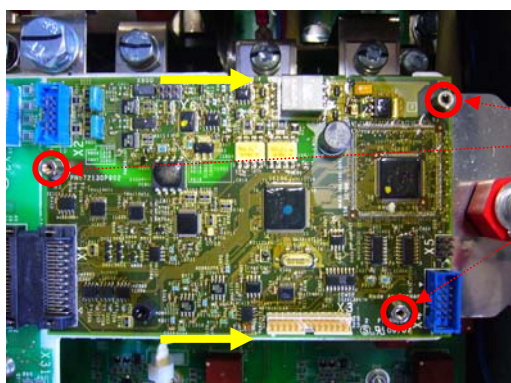
Size	Torque
M4x18	clips

Motor control Board: VX4A61101Y



Remove the 3 screws. (P1)

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on measuring Board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs. (P2)

Be careful, at the time of reassembling;
don't forget to put the steel.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

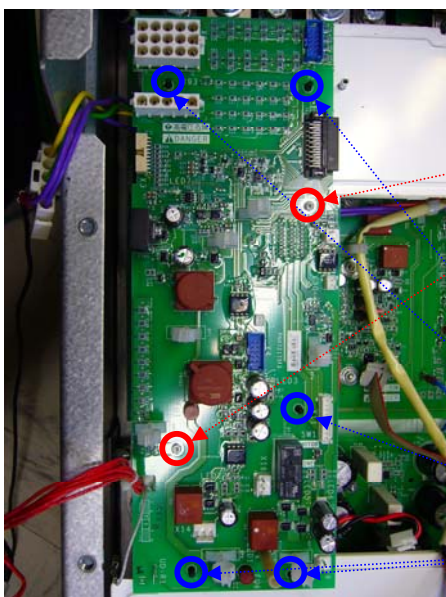
The motor control board can be removed.

Measuring Board: VX5A71HC13N4

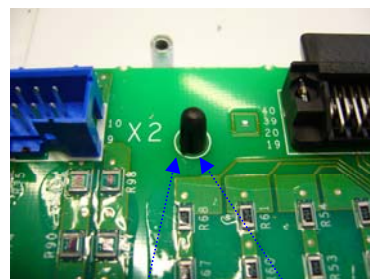


Disconnect the 13 wires, *top to the down*:

Ribbon X2-> X2 on motor control board
 Yellow/Green/Purple X6->shunt on power board
 Purple X93-> X2 on braking unit board
 Yellow/Green/Purple X5->alimentation Gate drive board IGBT Yellow/U Green/V Purple/W
 Ribbon X3 -> X31 on Gate drive board IGBT
 Black X31-> CN7A on soft charge board
 Red X30-> CN2A on Soft charge board
 White X21-> inside fan
 Ribbon X4-> X4 on control block
 Multicolour X11-> currents sensors U V W
 Yellow X14 -> X2 on Fan control board
 Ribbon X8->X82 on Power Board
 White UD-R1 -> discharging resistor



Remove the 2 screws.



Push on the 5 plastic supports to remove the board.

Push on the side.

The Measuring Board can be removed.



Size	Torque
M3x6	0.8Nm
M3x8	clips

DC Bus Bar kit: VZ3N1314



Remove 2 screws (P4)

Remove 1 screw (P2)

Remove 5 screws (P1)

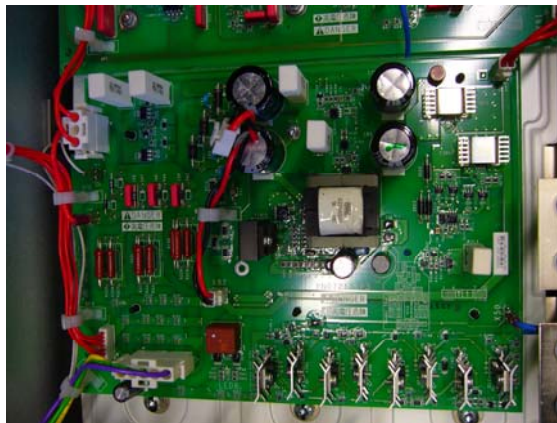
Remove 7 screws (P3)



Mark	Size	Torque
P1	M10x20	27Nm
P2	M6x12	5.5Nm
P3	M8x20	13.5Nm
P4		

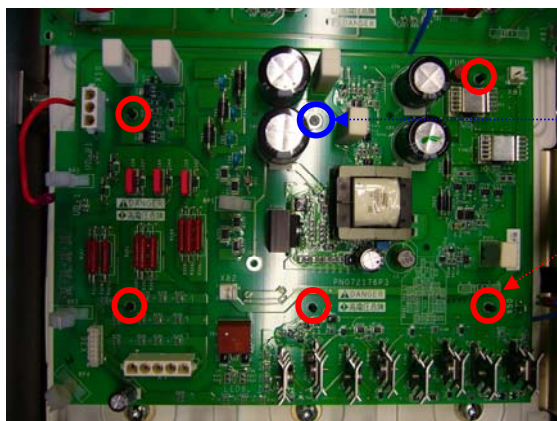
The DC bus bar kit can be removed.

Power board: VX5A1HC1316



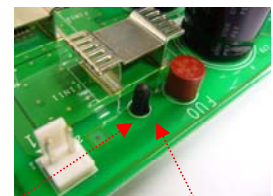
Disconnect the 8 wires, *top to the down and right to left:*

Red X81 -> X81 on Gate drive board IGBT
 Blue X50->DC bus Bar -> X51 on gate drive board IGBT
 Black/Red X82 ->X8 on Power Board
 Red X10-> X10 X11 on rectifier snubber circuit
 White UD+R1 -> Discharging resistor
 Red UD+1-> Power bus on U
 Red X30-> CN2A on Soft charge board
 Yellow/Green/Purple X7-> X11 Yellow X12 Green X13 Purple on filter card



Remove 1 screw

Push on the 5 plastic supports
to remove the board.



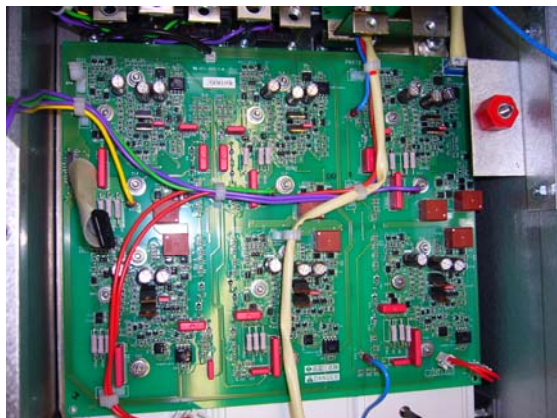
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x22	clips

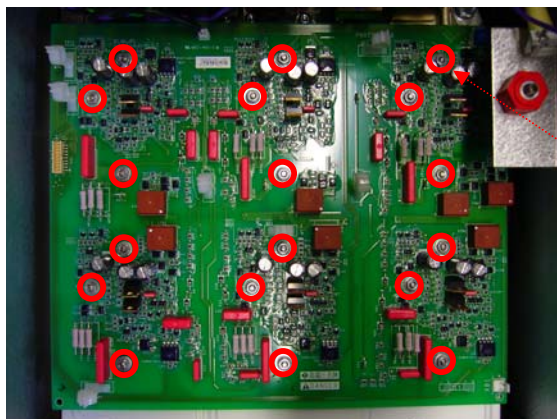
The Power Board can be removed.

Gate Drive Board IGBT: VX5A1200



Disconnect the 4 wires top to down and right to left.

Ribbon X91-> X1 on braking unit board
 Red X81 -> X81 on power Board
 Blue X52->X5 on braking unit board
 Purple X23-> X5 on measuring Board
 Blue X51-> DC bus Bar -> X50 on Power board
 Green X22-> X5 on measuring Board
 X4-> thermal sensor
 Yellow X21-> X5 on measuring Board
 Ribbon X31-> X3 on measuring board



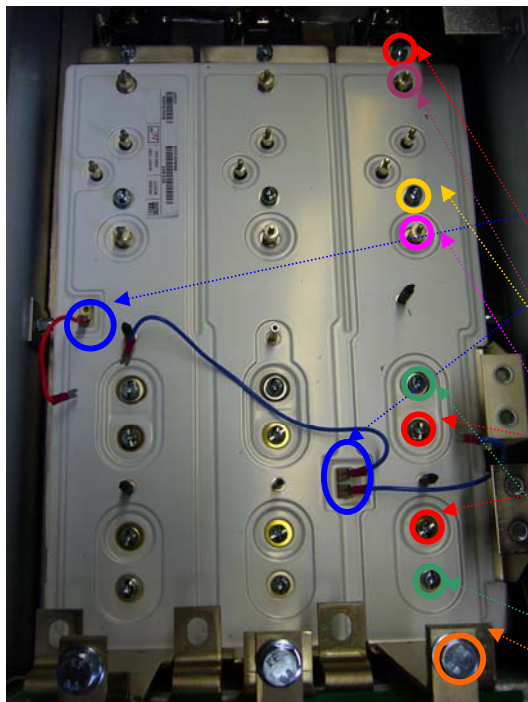
Remove 18 nuts

The Gate Drive Board IGBT can be removed.



Size	Torque
M4	1.2Nm

Bus bar-Phase Size 11: VZ3N1314



Disconnect the 4 wires Left to right.

Red wire on interconnection bar U -> UD+1 on power board
Blue wire on interconnection bar -> X51 on Gate Drive Board IGBT W
-> X50 on power Board

For each 3 AC bus bar, you should remove:

Remove 3 screws (P1-P2)

Remove 1 screw (P7)

Remove 1 screw (P6)

Remove 1 screw (P5)

Remove 2 screws (P3-P4)

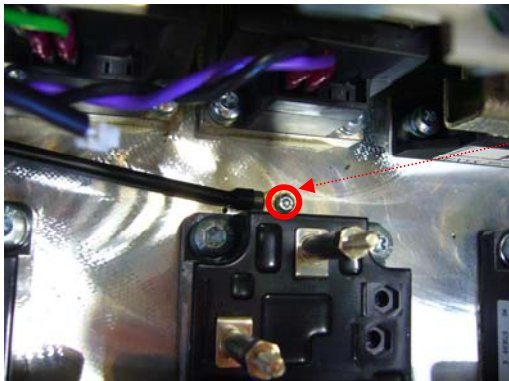
Remove 1 screw (P8)



Mark	Size	Torque
P1-P2	M6x12	3.3Nm
P6	ST 10x6	3.3Nm
P7	M6x20	3.3Nm
P5	ST 10X11	3.3Nm
P3-P4	M6x14	3.3Nm
P8	M12x25	45Nm

The Bus bar-Phase can be removed.

Thermal Sensor: VZ3G1104



Remove 1 screw.

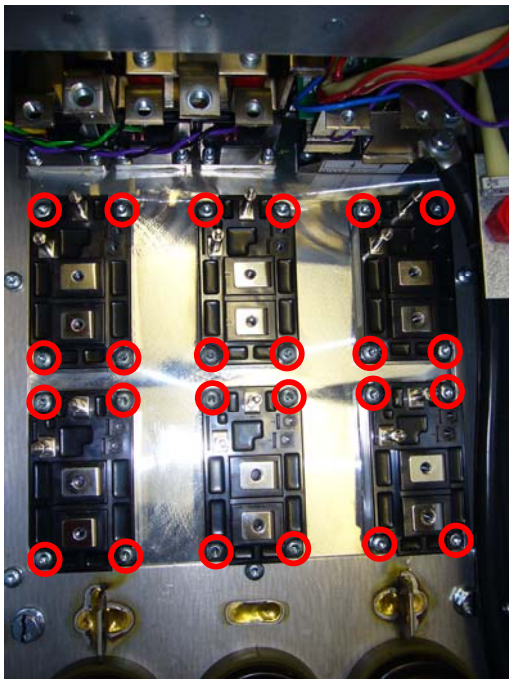
Be careful, don't forget to connect the wire on the connector "X4" of the board Braking unit kit. And "X4" of the Gate drive board IGBT for the over one

The thermal sensor can be removed.



Size	Torque
M3x6	0.8Nm

Module IGBT: VZ3IM1600M1271



Remove 24 screws (S?).

It is **not necessary** to change all IGBT but just this one that is damaged.

After changing, **be careful** to set up direction.

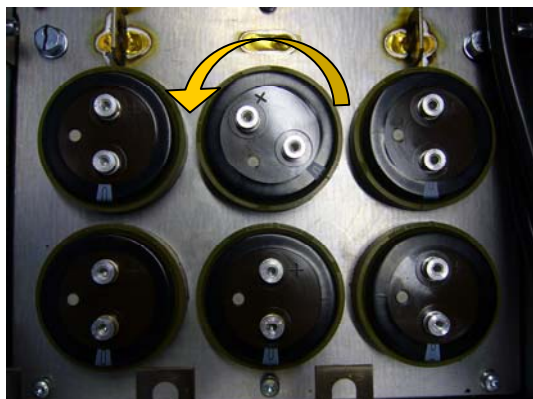
Don't forget applying the grease.

The IGBT can be removed.



Size	Torque
M6x20	3.0Nm

Lots of 6 capacitors: VY1ADC1112



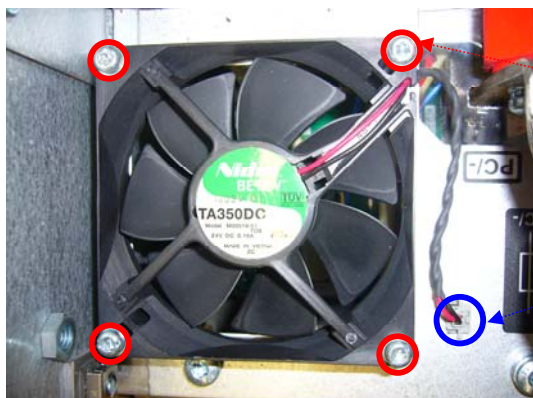
To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction



The Capacitors can be removed.

Internal fan 24 VDC: VZ3V1213



Remove 4 screws.

Disconnect 1 wire:

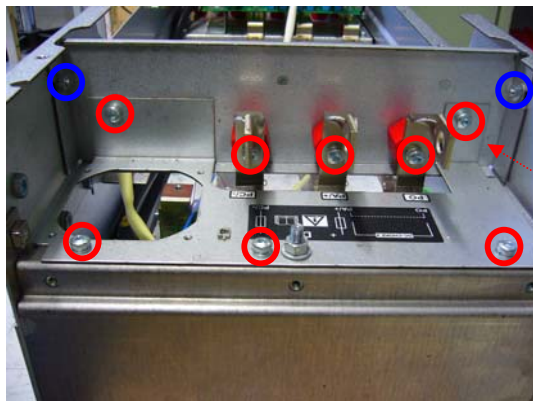
Black-> connector ->X21 on measuring board

The Internal Fan can be removed.



Size	Torque
M4x35	1.2Nm

Metal Body Parts KIT



Remove 2 nuts (P2).

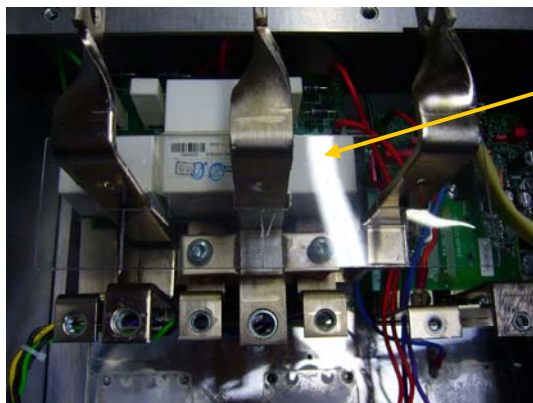
Remove 8 screws (P1)



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm

The Metal Body Parts KIT can be removed.

Plastic Parts Kit: VY1A1401

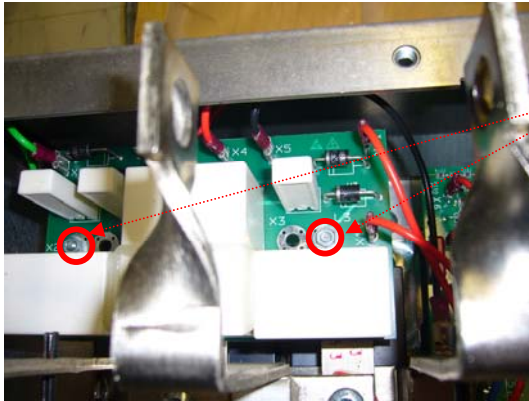


Remove Plastic part

Don't forget to replace it during reassembling

The Plastic Part KIT can be removed.

Rectifier Snubber Circuit: VX4A1200



Remove 2 nuts.

Disconnect 3 Wires left to right

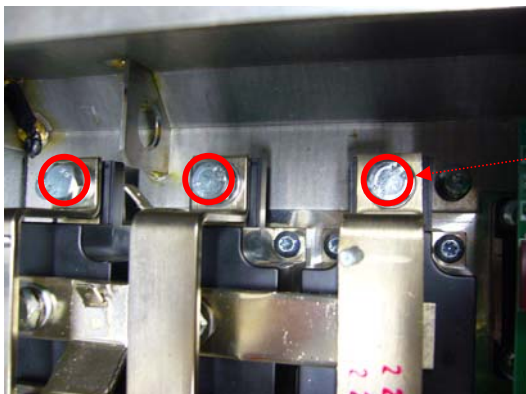
Green X1 -> bus bar L2
Red X4 -> bus bar PO
Black X5 -> bus bar PC

The Rectifier snubber circuit can be removed.



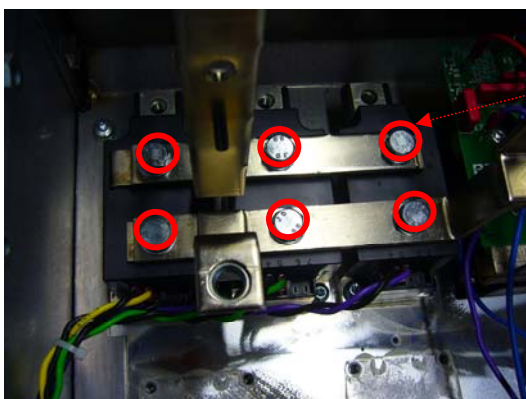
Size	Torque
M4	1.2Nm

Rectifier Module: VZ3TD1285M1671 and VZ3N1315



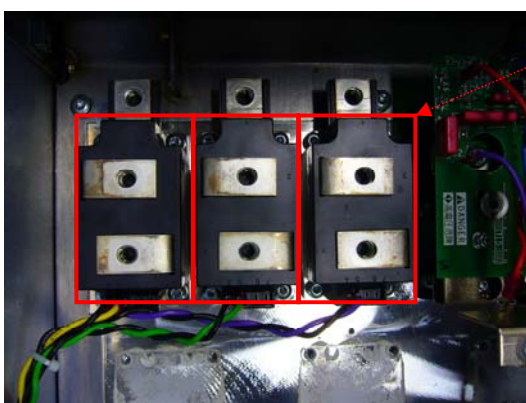
Part 1

Remove 3 screws (P1)



Part 2

Remove 6 screws (P1)



Part 3

Remove 12 screws (P2)

It is **not necessary** to change all rectifiers but just this one that is damaged.

After changing, **be careful** to set up direction.
Don't forget applying the grease.

The Rectifier can be removed.



Mark	Size	Torque
P1	M10x25	13.5Nm
P2	M5x25	5.0Nm

Rectifier Module: VZ3TD1285M1671

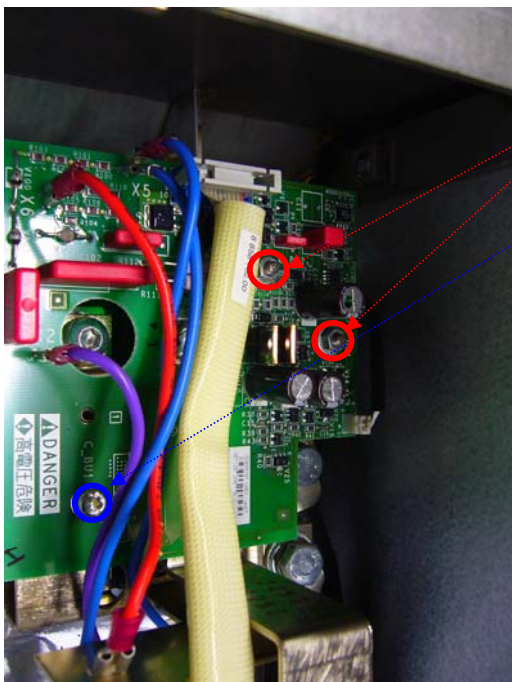


Be careful, after changing respect the colour line for the wires.

From left to right
[Black - Yellow]- [Black - Green] - [Black - Purple]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Braking Unit kit: VZ3F1110



Part 1

Remove 2 nuts (P1).

Remove 1 screw (P2)

Disconnect 5 Wires top to down

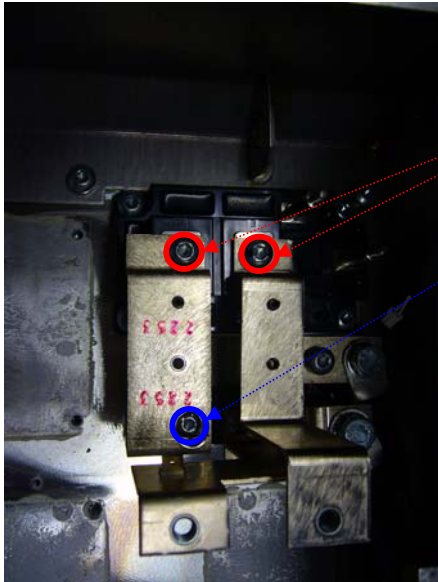
Ribbon X1 -> X91 on gate drive board IGBT
Blue X5 -> X52 on gate drive board IGBT and DC bus bar
Red X6 -> bus bar PA
Purple X2 -> X93 on measuring board
Black X4 -> thermal sensor

The Braking Unit KIT can be removed.



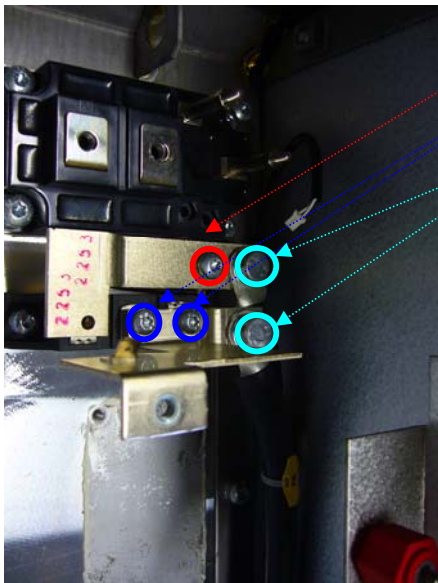
Mark	Size	Torque
P1	M4	1.2Nm
P2	M3x6	0.8Nm

Braking Unit kit: VZ3F1110



Part 2
Remove 2 screws (P1).

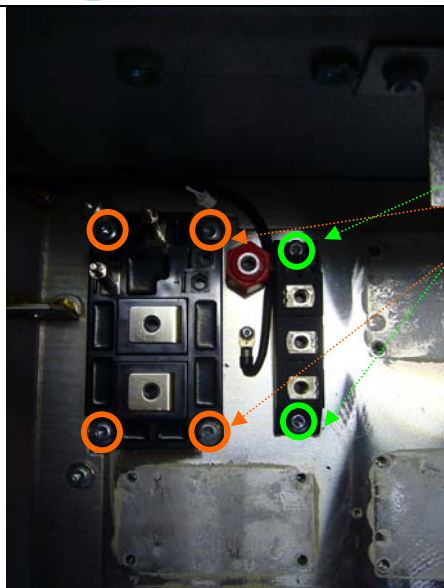
Remove 1 screw (P2)



Part 3
Remove 1 screw (P1)

Remove 2 screws (P2)

Remove 2 screws (P3)



Part 4

Remove 2 screws (P4)

Remove 4 screws (P5)

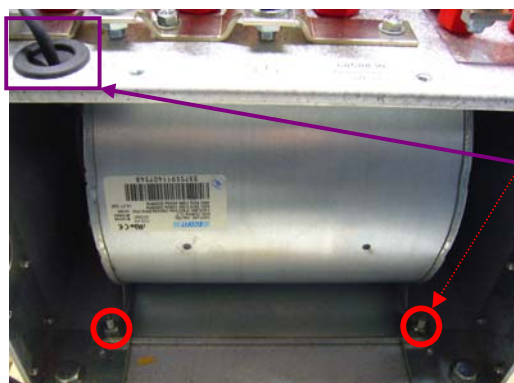
After changing, **be careful** to set up direction.
Don't forget applying the grease.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6x14	5.5Nm
P3	M8x20	13.5Nm
P4	M6x20	3.0Nm
P5	M5x25	5.0Nm

The Braking Unit KIT can be removed.

Fan Power Electronic. VZ3V1212



Remove 2 nuts

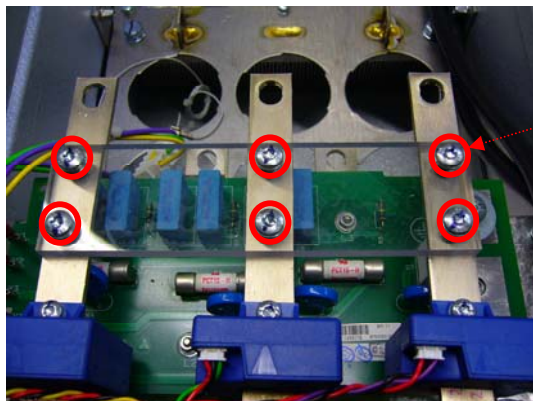
Be careful, to remove the fan, remove the gasket and slide the wire in the hole.



Size	Torque
M6	5.5Nm

The Fan Power electronic can be removed.

Plastic Parts Kit: VY1A1401



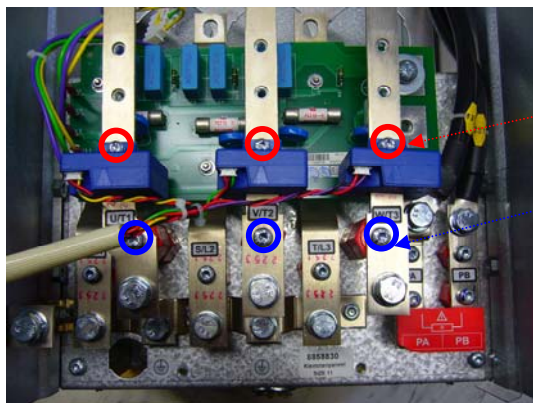
Remove 6 screws

The Plastic Part KIT can be removed.



Size	Torque
M6x20	5.5Nm

Motor Current Sensor: VY1A1107



Remove 3 screws (P1)

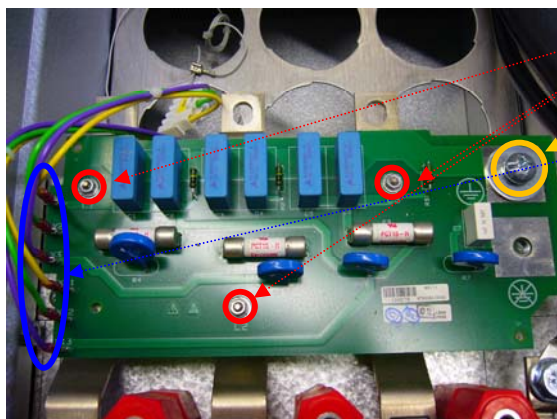
Remove 3 screws (P2)

The Motor Current Sensor can be removed.



Mark	Size	Torque
P1	M4x12	1.2Nm
P2	M6x14	5.5Nm

RFI Filter Board: VX4A1113



Remove 3 nuts (P1)

Remove 1 screw (P2)

Disconnect 6 wires top to down:

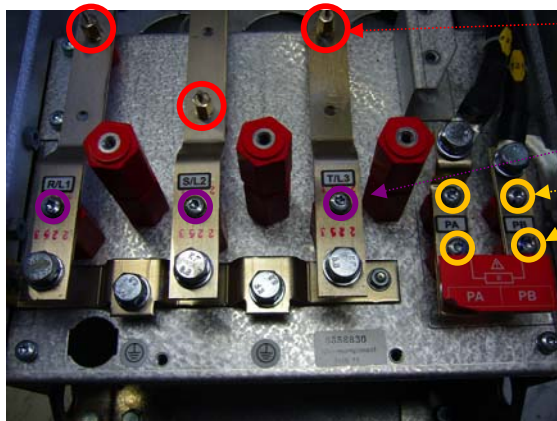
Yellow X1 -> X1 on fan control Board
 Green X2 -> X1 on fan control Board
 Purple X3 -> X1 on fan control Board
 Yellow X11 -> X7 on power Board
 Green X12 -> X7 on power Board
 Purple X13 -> X7 on power Board

The RFI filter board can be removed.



Mark	Size	Torque
P1	M4	1.2Nm
P2	M6x12	5.5Nm

AC Bus Bar Kit: VZ3N1315



Remove 3 screws (P1)

Remove 3 screws (P2)

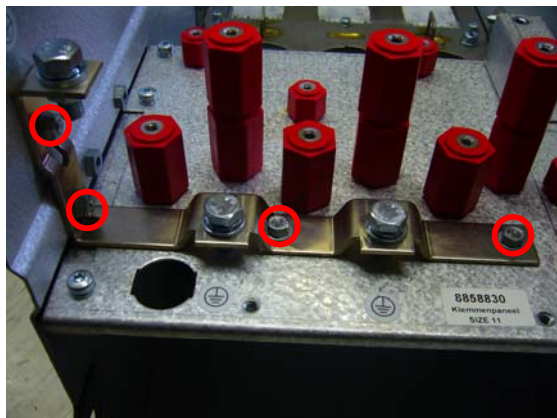
Remove 4 screws (P3)

The AC Bus bar can be removed.



Mark	Size	Torque
P1	ST 10x11	5.5Nm
P2	M6x14	5.5Nm
P3	M6x12	5.5Nm

Earth Terminal Bar: VZ3N1315



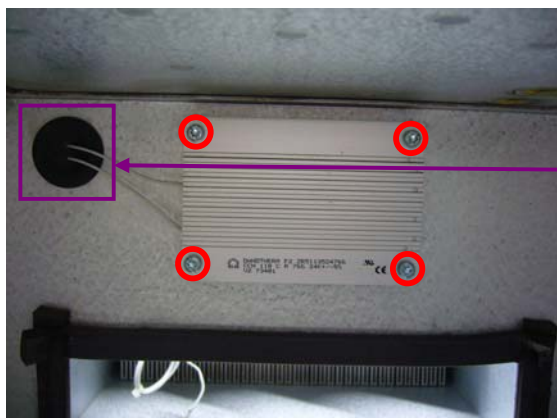
Remove 4 nuts

The Earth Terminal Bar can be removed.



Size	Torque
M6	5.5Nm

Discharging Resistor. VZ3R24KW125



Remove 4 screws (S?)

Be careful, to remove the Discharging resistor, remove the gasket and slide the wire in the hole.

Discharging resistor can be removed.



Size	Torque
M4x20	1.2Nm

DC Choke: VW3A4520



Remove the 4 screws(S?).



Size	Torque
M6	5.5Nm

The DC Choke can be removed.



9.23.2 Product Assembling Drawing

No information

9.23.3 Product Cabling Drawing

Refer to following file: [Cabling diagram size 11.pdf](#)

9.24 ATV61/71 Size 12 (size, refer to 1.2)

9.24.1 Dismantling and reassembling

Size 12: ATV71HC16N4, ATV61HC22N4

ATV71HC16N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1322	DC bus bar kit
VZ3N1319	AC Bus Bar KIT
VZ3N1317	Wires KIT
VZ3IM1402M1271	Module IGBT pired (400A / 1200V)
VZ3G1104	Thermal Sensor
VZ3F1110	Braking Unit KIT
VY1ADV1108	Screws KIT
VY1ADC1113	Lot de 8 capacitors (5200µF / 400V)
VY1A1402	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1213	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A1HC1622	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1201	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1114	RFI Filter Board
VW3A4521	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HC22N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TD1330M1601	Rectifier Module (Thyristor / Diode) 330A / 1600V
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1322	DC Bus Bar KIT
VZ3N1319	AC Bus Bar KIT
VZ3N1317	Wires KIT
VZ3IM1402M1271	Module IGBT pired (400A / 1200V)
VZ3G1104	Thermal Sensor
VZ3F1110	Braking Unit KIT
VY1ADV1108	Screws KIT
VY1ADC1113	Lot de 8 capacitors (5200µF / 400V)
VY1A1402	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1213	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A1HC1622	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1201	Gate Drive Board
VX4A61101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1114	RFI Filter Board
VW3A4521	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Cover with I/O Terminal Cover: VY1A1213



Remove 9 screws.



The front cover can be removed.



Size	Torque
M6x12	5.5Nm

Plastic Parts KIT: VY1A1402



Push here and disengage plastic part

Remove 2 nuts

The Plastic part KIT can be removed.



Size	Torque
M6	5.5Nm

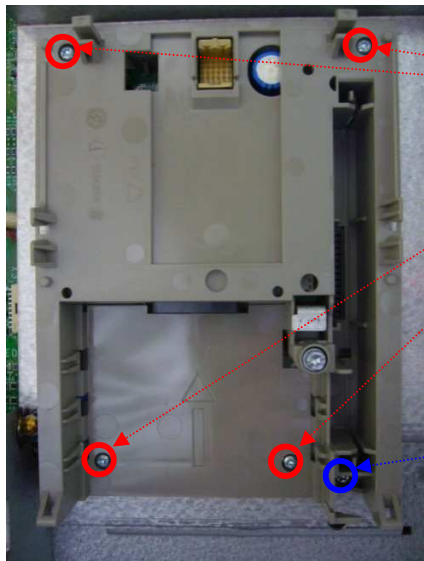
Control block: VX4A61101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.

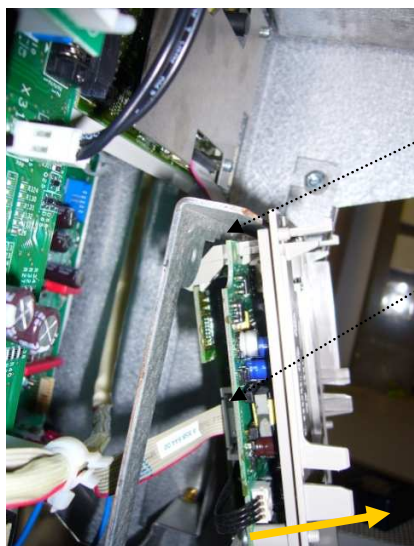


Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 to S8).

Remove 1 screw (S38).



Disconnect the ribbon cable.

X3 from Control bloc "interface Board"->X3 on Motor Control Board

Disconnect the ribbon cable.

X4 from Control bloc "interface Board"->X4 on Power Board

The Control block can be removed.

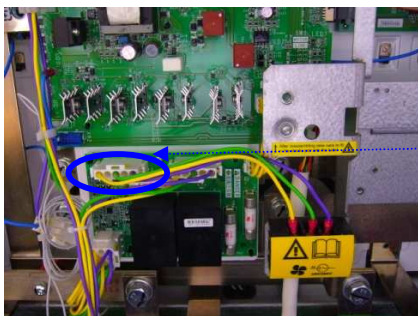


Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5NM

Metal Support for Control bloc



Remove 2 nuts.



Disconnect the wire:

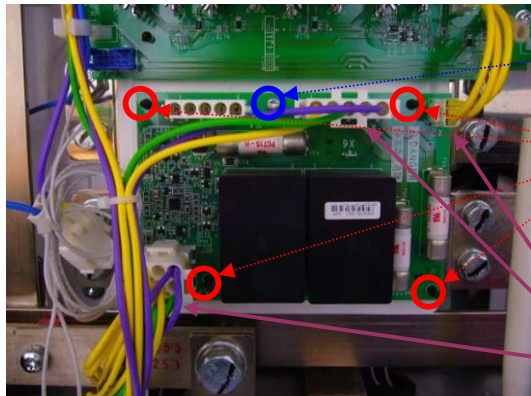
Yellow/Green/Purple -> X4 on fan control board

The metal support can be removed.



Size	Torque
M6	5.5Nm

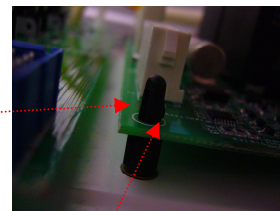
Fan Control Board: VX5A1400



Remove 1 screw.

Push on the 4 plastic supports to remove the board.

Disconnect the wires. From right to left:
 Yellow X2->X14 on power board
 Yellow/Green/Purple X1-> X1 Yellow X2 Green X3 Purple on filter card
 Yellow/Green/Purple X3-> blower power



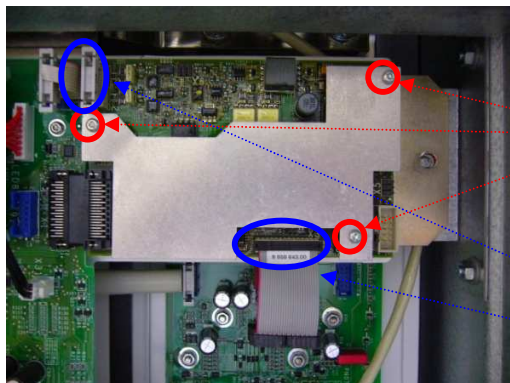
Push on the side.

The Fan control board can be removed.



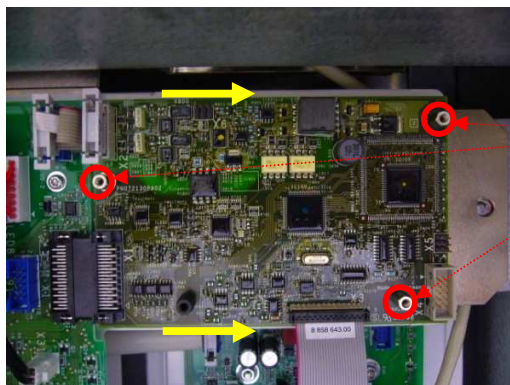
Size	Torque
M3x6	0.8Nm
M3x8	clips

Motor control Board: VX4A61101Y



Remove the 3 screws. (P1)

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on Power Board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs. (P2)

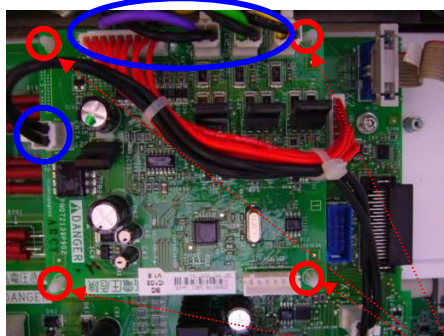
Be careful, at the time of reassembling;
don't forget to put the steel.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

The motor control board can be removed.

Soft Charge board: VX5A1300



Disconnect the 5 wires, *left to the right*:

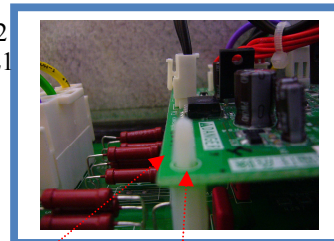
Black CN7A-> X31 on Power Board

Red CN2A-> X30 on Power Board

Purple/Black CNL3G ->Gate rectifier 3 on I3

Green/Black CNL2G ->Gate rectifier 2 on L2

Yellow/Black CNL1G -> Gate rectifier 1 on L1



Push on the 4 plastic supports
to remove the board.

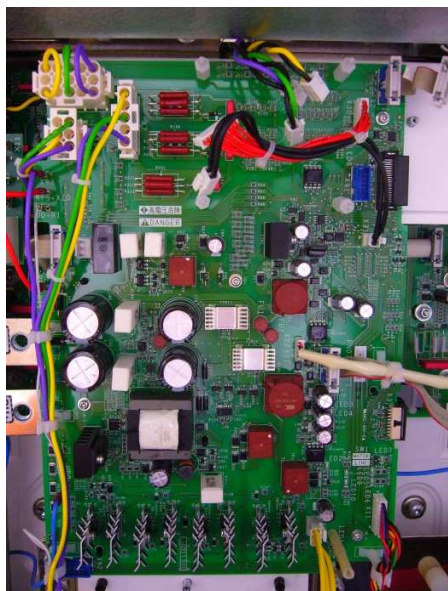
Push on the side.

The Soft charge Board can be removed.



Size	Torque
M3x8	clips

Power board: VX5A1HC1622



Disconnect the 19 wires, *top to the down*:

Ribbon X2-> X2 on motor control board

Red X30-> CN2A on Soft charge board

Black X31-> CN7A on soft charge board

Ribbon X91->X1 on braking unit board

Purple X93->X2 on braking unit board

Yellow/Green/Purple X7-> X11 Yellow X12 Green X13 Purple on filter card

Yellow/Green/Purple X6->shunt on power board

Yellow/Green/Purple X5->alimention Gate drive board IGBT Yellow/U Green/V Purple/W

Red RFS+X10 ->X10 on Rectifier Snubber Circuit.

White UD+R1 -> Discharging resistor

Ribbon X8->X8 on Command IGBT card U

Red RFS+X11 ->X11 on Rectifier Snubber Circuit

Blue UD0 -> Power Interconnection bar U

White UD-R1 -> discharging resistor

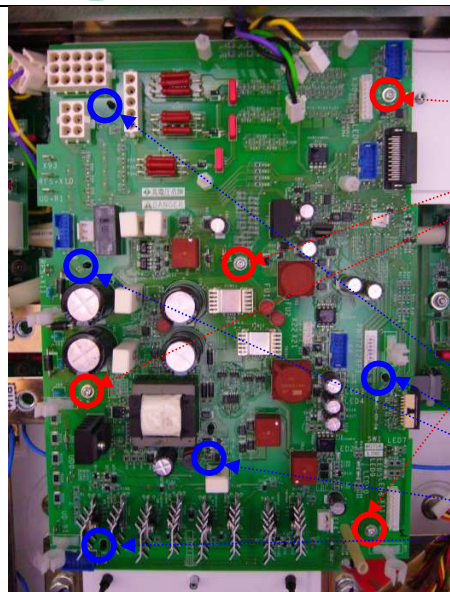
White X21-> inside fan

Ribbon X4-> X4 on control block

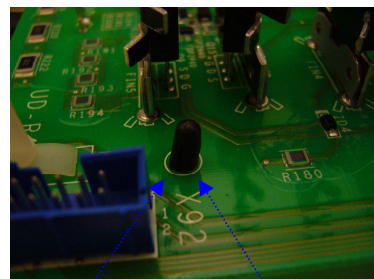
Ribbon X3 -> X31 on command IGBT board W

Yellow X14 -> X2 on Fan control board

Multicolor X11-> currents sensors U/V/W



Remove the 4 screws.



Push on the 5 plastic supports to remove the board.

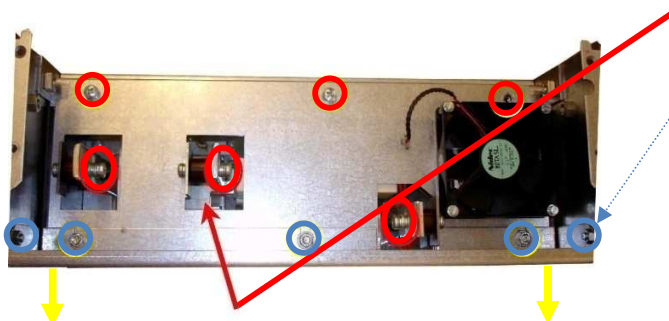
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

The Power Board can be removed.

Internal Fan 24 VDC: VZ3V1213



Remove 6 screws (P1).

Remove 5 nuts (P2).

The Internal fan can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm